

Tensor-Based Joint Channel Estimation and Symbol Detection for Time-Varying mmWave Massive MIMO Systems

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Abstract—In this paper, a tensor-based joint channel parameter estimation and information symbol detection scheme is developed for millimeter wave (mmWave) massive multiple-input multiple-output (MIMO) communication systems. At the base station (BS), the information symbols are encoded according to the Khatri-Rao space-time (KRST) method and transmitted through time-varying channels. The received signals at the mobile station (MS) are constructed into a nested complex-valued parallel factor (PARAFAC) model, which contains an outer model and an inner model, respectively. With outer model, we estimate the compound channel matrix and detect the information symbols considering the sparse scattering nature of mmWave channels. With inner model, we extract physical parameters, including angles of arrival/departure (AoAs/AoDs), Doppler shifts and complex path gains from the estimated compound channel matrix. These physical parameters can be used to significantly reduce feedback overhead. A tricky way here is that we convert complex inner-submodel into a real one such that the computational complexity is reduced. Compared with existing schemes, the proposed one improves the estimation accuracy with low computational complexity, and is applicable for both uniform linear arrays (ULAs) and uniform planar arrays (UPAs). Simulation results demonstrate the effectiveness of the proposed joint channel estimation and information symbol detection scheme.

Index Terms—Tensor, joint estimation and detection, mmWave, massive MIMO, time-varying channels.

I. INTRODUCTION

MILLIMETER wave (mmWave) communication systems cover the frequency range from 26.5 GHz to 300 GHz, and can provide large bandwidth to meet the requirements of

extremely high information transmission rate [1], [2]. Meanwhile, the deployment of a large number of antennas at the base station (BS) can improve the spatial resolution, and focus the beam energy within a precise range, thereby effectively eliminating multi-user interference and reducing transmission power waste. Due to the shorter wavelength of mmWave, large-scale antenna arrays are allowed to be configured at the same physical dimension, which helps further explore spatial multiplexing capabilities and significantly improves spectral and energy efficiency [3], [4]. Therefore, the integration of mmWave and massive multiple-input multiple-output (MIMO) (i.e., mmWave massive MIMO technique) has become a key technology in 5G and even future 6G.

Mmwave massive MIMO systems rely on the perfect channel state information (CSI) that is usually unknown and needs to be obtained through channel estimation for practical communication systems. Traditional schemes execute channel estimation at receivers through compressed sensing (CS) algorithm. In [5], an adaptive CS (ACS) method is developed to estimate CSI, where the training process is divided into several stages, and training codebooks designed for each stage are based on the results of the previous stage. Though the ACS method has more effective performance than conventional CS solutions, its codebook size places restrictions on the resolution of the estimated channel. In [5], the continuous parameter space is discretized into a finite set of grid points, which inevitably incurs grid mismatch. The work [6] proposes a super-resolution (SR) CS-based method for joint dictionary parameter training and sparse signal recovery. A weighted sum function between sparsity and data fitting error is first constructed by taking advantage of the sparsity of the solution. Then an iterative reweighted approach is designed to refine the unknown parameters and sparse signals. The SR method overcomes the grid mismatch and extracts parameters with high precision.

However, the CSI estimation schemes in [5] and [6] need to frequently send pilot sequences when the CSI varies rapidly, which leads to a severe reduction in spectral efficiency. How to alleviate pilot overhead is of great importance for mmWave massive MIMO systems. Actually, the dimensions of tensors can be associated with several signal dimensions such as space, time and frequency. Different from the traditional matrices, the low-rank decomposition of tensors is unique under appropriate

Manuscript received April 11, 2021; revised July 28, 2021, September 15, 2021, and October 9, 2021; accepted October 29, 2021. Date of publication November 8, 2021; date of current version November 24, 2021. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. An Liu. This work was supported in part by the National Natural Science Foundation of China under Grants 61831013, 61601414, and 62171416, in part by the Fundamental Research Funds for the Central Universities under Grant CUC210B005, and in part by Tsinghua University-China Mobile Communications Group Company Ltd. Joint Institute. (Corresponding author: Feifei Gao.)

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Digital Object Identifier 10.1109/TSP.2021.3125607

conditions. Hence tensor decomposition can be used in wireless communications for blind or semi-blind identification of system parameters [7]–[14].

Due to the multi-dimensionality and sparsity of mmWave massive MIMO channels, the tensor low-rank decomposition techniques can be fully exploited to obtain higher channel estimation accuracy with less pilot overhead. For mmWave MIMO orthogonal frequency division multiplexing (MIMO-OFDM) systems, a parallel factor (PARAFAC) tensor decomposition based downlink channel estimation is developed in [7], and the corresponding Cramér-Rao bound (CRB) is also derived. In [8], a tensor-based joint channel parameter estimation method is proposed for dual-polarized mmWave massive MIMO systems, which greatly relieves the pilot overhead. Considering the characteristic that angles of arrival/departure (AoAs/AoDs) vary more slowly in mmWave channels, the work [9] proposes a two-stage channel estimation method with the PARAFAC model. For the same pilot size, the method in [9] has higher channel estimation accuracy than traditional CS-based methods. However, only third-order tensor models are adopted by the aforementioned works [7]–[9], which fails to make full use of rich dimension resources of mmWave massive MIMO systems.

The Khatri-Rao space-time (KRST) coding method developed in [15] has received considerable attention since it is applicable to design space-time codes with full-diversity or full-rate, or variable rate-diversity trade-off and built-in channel blind identifiability. In [10], a KRST-based space-time-frequency coding scheme is presented for MIMO-OFDM systems. The received signals are formulated as a fourth-order nested PARAFAC tensor model, and a semi-blind receiver is derived for joint channel estimation and symbol detection. The work [11] considers a non-regenerative cooperative MIMO system without the direct link from source to destination. By exploiting the KRST at source and relay, the signals received at destination can also be constructed into a nested PARAFAC model. Different from [11], the work [12] formulates the received signals as a nested PARAFAC model by combining source-destination and source-relay-destination links to further improve the symbol detection performance. Besides, the receiver in [12] performs well under both blind and semi-blind configurations. More recently, two closed-form semi-blind receivers are presented in [13] by constructing a block Tucker-2 model at the relay and a Tucker-2 model at the sources. For a MIMO communication system in a time-varying multipath environment, [14] proposes a semi-blind receiver to jointly estimate the information symbols and multipath parameters. By combining KRST and multiple Khatri-Rao product of symbol matrices (MKRSM) codings, the received signals are constructed into a generalized nested PARAFAC model. However, the approaches in [10]–[14] are suitable for point-to-point MIMO or MIMO relay systems, and inevitably suffer from high computational complexity if being directly applied to massive MIMO systems, due to large number of antennas.

In this paper, by exploiting the KRST coding and antenna array structure characteristics, we design an optimized nested PARAFAC (ONP) tensor model for mmWave massive MIMO

time-varying channels. Then an improved two-stage alternating least squares (ITS-ALS) algorithm is proposed for joint channel parameter estimation and information symbol detection. The main contributions of this paper are summarized as follows:

- By resorting to the multi-dimensionality of mmWave massive MIMO time-varying channels and the extra time diversity of KRST coding, the received signals are constructed into a fourth-order tensor that can satisfy the nested PARAFAC model, which has outer PARAFAC model and inner PARAFAC model. On the basis of the sparse scattering nature of mmWave channels, the bilinear ALS (BALS) method is optimized by the singular value projection (SVP) algorithm to fit the outer model. By using the forward-backward and unitary transformation technique, the inner model is converted into a real-valued one. Then the optimized real-valued (OR) trilinear ALS (TALS) algorithm is proposed to accelerate the fitting of the inner PARAFAC model.
- Thanks to the multilinear algebraic structure of the nested PARAFAC model, the proposed ONP-ITS-ALS algorithm can obtain accurate CSI for time-varying channels. Besides, channel parameters, including AoAs, AoDs, Doppler shifts and complex path gains at each time block, can be extracted. Unlike the existing pilot-assisted estimation methods [5], [6], [16]–[18], the proposed ONP-ITS-ALS algorithm does not need to transmit channel training sequences, and thus saves spectral resources. Compared with traditional nested PARAFAC (TNP) based estimation schemes [10]–[12], the proposed approach achieves higher estimation accuracy with substantial computational complexity reduction, and is applicable for both uniform linear arrays (ULAs) and uniform planar arrays (UPAs).
- A theoretical study on the uniqueness issues and computational complexity is carried out. According to analysis and simulations, the proposed ONP-ITS-ALS algorithm can jointly estimate channel parameters and detect information symbols uniquely with more relaxed conditions. We can also flexibly choose parameters to save resources and improve estimation performance. Moreover, the proposed ONP-ITS-ALS algorithm works robustly across different scattering paths.

The rest of this paper is organized as follows. In Section II, the time-varying mmWave channel and nested PARAFAC signal models are introduced. The ONP-ITS-ALS algorithm is deduced in Section III. Uniqueness issues and computational complexity are discussed in Section IV. Simulation results are demonstrated in Section V, and conclusions are drawn in Section VI.

Notations and preliminaries: $\mathbb{R}$ and $\mathbb{C}$ are used to denote the Real and Complex domain, respectively. Letters $(a, b, \dots, A, B, \dots)$, boldface lower-case letters $(\mathbf{a}, \mathbf{b}, \dots)$, boldface capitals $(\mathbf{A}, \mathbf{B}, \dots)$, and calligraphy letters $(\mathcal{A}, \mathcal{B}, \dots)$ represent scalars, vectors, matrices, and tensors, respectively. By introducing the matrix symbol $[\cdot]$ and dual-matrix symbol $[\![\cdot]\!]$, the matrix $\mathbf{A}$ and tensor $\mathcal{A}$ can be respectively expressed as $\mathbf{A} = [a_{m,n}]_{m,n=1}^{M,N}$ and $\mathcal{A} = [\![a_{i_1, \dots, i_N}]\!]_{i_1, \dots, i_N=1}^{I_1, \dots, I_N}$. The Euclidean norm and Frobenius norm are denoted by $\|\cdot\|_2$

and $\|\cdot\|_F$, respectively. The transpose, conjugate, conjugate transpose, Moore-Penrose pseudo-inverse, the m -th row and the n -th column of $\mathbf{A} \in \mathbb{C}^{M \times N}$ are represented by $\mathbf{A}^T$, $\mathbf{A}^*$, $\mathbf{A}^H$, $\mathbf{A}^\dagger$, $\mathbf{A}(m, :)$, and $\mathbf{A}(:, n)$, respectively. $r(\mathbf{A})$ and $k(\mathbf{A})$ indicate the rank and Kruskal rank (k -rank) of $\mathbf{A} \in \mathbb{C}^{M \times N}$. $D_m\{\mathbf{A}\}$ stands for a diagonal matrix formed by $\mathbf{A}(m, :)$. $\text{vec}(\mathbf{A})$ denotes the vector obtained by stacking $\mathbf{A}(:, n)$ for $n = 1, \dots, N$. The $N \times N$ identity matrix is denoted by $\mathbf{I}_N$. Similarly, the identity N -order tensor with dimensions $I \times I \times \dots \times I$ is denoted by $\mathcal{I}_{N,I}$ with elements 1 for $i_1 = \dots = i_N \in \{1, \dots, I\}$. Π_N stands for an $N \times N$ exchange matrix with all unities on its secondary diagonal and zeros elsewhere. Notations $\otimes$ and $\odot$ denote the Kronecker product and Khatri-Rao product. They enjoy the following property:

$$\begin{aligned} (\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) &= (\mathbf{AC}) \otimes (\mathbf{BD}), \\ (\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \odot \mathbf{D}) &= (\mathbf{AC}) \odot (\mathbf{BD}), \end{aligned} \quad (1)$$

where matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are of appropriate size for mathematical operations.

The PARAFAC model of an N -th order tensor $\mathcal{Z} \in \mathbb{C}^{I_1 \times I_2 \times \dots \times I_N}$ can be described by the element-wise format as

$$z_{i_1, \dots, i_N} = \sum_{k=1}^K \prod_{n=1}^N a_{i_n, k}^{(n)} \quad (2)$$

with equivalently mode- n product

$$\mathcal{Z} = \mathcal{I}_{N,K} \times_{n=1}^N \mathbf{A}^{(n)}, \quad (3)$$

in which $\mathcal{Z} = \llbracket z_{i_1, \dots, i_N} \rrbracket_{i_1, \dots, i_N=1}^{I_1, \dots, I_N}$, $\mathbf{A}^{(n)} = [a_{i_n, k}^{(n)}]_{i_n=1}^{I_n, K}$ for $n = 1, \dots, N$. The achievable minimum K is called the rank of $\mathcal{Z}$. Tensors are often reorganized into matrices for analysis and calculation, which is known as matricizing or unfolding. The mode- n unfolding form of $\mathcal{Z}$ can be expressed as

$$[\mathcal{Z}]_n = (\mathbf{A}^{(N)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)}) \mathbf{A}^{(n)T} \in \mathbb{C}^{I_1 \dots I_{n-1} I_{n+1} \dots I_N \times I_n}, \quad (4)$$

where $[\mathcal{Z}]_n = [z_{i_1, \dots, i_{n-1}, i_{n+1}, \dots, i_N, i_n}^{(n)}]_{i_1, \dots, i_{n-1}, i_{n+1}, \dots, i_N, i_n=1}^{I_1, \dots, I_{n-1}, I_{n+1}, \dots, I_N, I_n}$ with correspondence $z_{i_1, \dots, i_N} = z_{j, i_n}^{(n)}$ and $j = 1 + \sum_{m=1, m \neq n}^N [(i_m - 1) \prod_{l=1, l \neq n}^{m-1} I_l]$.

II. SYSTEM MODEL

A. Time-Varying Channel Model

We consider a mmWave massive MIMO system, where the BS has M_T transmit antennas and M_{RF} radio frequency (RF) chains, while the mobile station (MS) is equipped with N_R receive antennas and N_{RF} RF chains. Normally, the number of RF chains is less than that of antennas, i.e., $M_{RF} < M_T$ and $N_{RF} < N_R$. We commence with the typically adopted point scatterer case and assume that there are K scatter paths between BS and MS. For mmWave communication systems, the signals usually suffer from severe path loss, and their reflection to the surrounding environment tends to be less [19]–[21]. Therefore, the value of K is generally small. At the n -th time block, the time-varying

mmWave channel can be modeled as [20]–[22]

$$\mathbf{H}[n] = \sum_{k=1}^K \alpha_{k,n} e^{j2\pi f_d^k n T_s} \mathbf{a}_R(\theta_{k,n}^{\text{az}}, \theta_{k,n}^{\text{el}}) \mathbf{a}_T^H(\phi_{k,n}^{\text{az}}, \phi_{k,n}^{\text{el}}), \quad (5)$$

where $\alpha_{k,n}$ is the complex time-varying gain following the complex Gaussian distribution with zero mean and unit variance, f_d^k denotes the Doppler shift of the k -th scatter path, T_s is the sampling period, $\phi_{k,n}^{\text{az}}$ and $\phi_{k,n}^{\text{el}}$ are azimuth and elevation AoDs at the BS, $\theta_{k,n}^{\text{az}}$ and $\theta_{k,n}^{\text{el}}$ are azimuth and elevation AoAs at the MS. Moreover, $\mathbf{a}_T(\phi_{k,n}^{\text{az}}, \phi_{k,n}^{\text{el}})$ and $\mathbf{a}_R(\theta_{k,n}^{\text{az}}, \theta_{k,n}^{\text{el}})$ are the transmit and receive steering vectors, respectively. They are determined by the geometry of the antenna array at the BS and MS. For the simpler ULAs, the array steering vector can be determined by one angle. Ignoring the subscripts without loss of generality, it can be represented by [23]

$$\mathbf{a}_{\text{ULA}}(\psi) = [1, e^{-j2\pi d\psi/\lambda}, \dots, e^{-j2\pi d(M-1)\psi/\lambda}]^T, \quad (6)$$

where $\psi = \sin(\beta)$, β denotes the AoD ϕ or AoA θ , M is the number of transmit or receive antennas, λ denotes the signal wavelength, d is the distance between two adjacent antennas and is generally set to half the signal wavelength. For the UPAs with $M = M^x M^y$ antennas, where M^x and M^y are the number of elements in x-direction and y-direction, respectively, the distance between two adjacent antennas in x-direction and y-direction are assumed to be d_x and d_y . Define

$$\eta^x = \sin(\psi^{\text{el}}) \cos(\psi^{\text{az}}), \quad \eta^y = \sin(\psi^{\text{el}}) \sin(\psi^{\text{az}}), \quad (7)$$

where ψ^{az} and ψ^{el} denote azimuth and elevation AoD or AoA, respectively. The array steering vector can be given by [24]

$$\mathbf{a}_{\text{UPA}}(\psi^{\text{az}}, \psi^{\text{el}}) = \mathbf{a}_{\text{ULA}}(\eta^y) \otimes \mathbf{a}_{\text{ULA}}(\eta^x), \quad (8)$$

where $\mathbf{a}_{\text{ULA}}(\eta^x)$ and $\mathbf{a}_{\text{ULA}}(\eta^y)$ are defined by (6) with M replaced by M^x or M^y , ψ replaced by η^x or η^y , and d replaced by d_x or d_y . The steering matrix is then given by

$$\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_K] \in \mathbb{C}^{M \times K}, \quad (9)$$

where $\mathbf{a}_k$ denotes the steering vector of the k -th scatter path, which follows the form of (6) or (8).

Although the channel may change rapidly, the variation of complex path gains $\alpha_{k,n}$ for different time block is very slow. Moreover, $(\phi_{k,n}^{\text{az}}, \phi_{k,n}^{\text{el}})$ and $(\theta_{k,n}^{\text{az}}, \theta_{k,n}^{\text{el}})$ are only sensitive to the large-scale scattering environment and do not vary with small-scale mobility. Therefore, it is reasonable to assume that $\alpha_{k,n}$, $(\phi_{k,n}^{\text{az}}, \phi_{k,n}^{\text{el}})$ and $(\theta_{k,n}^{\text{az}}, \theta_{k,n}^{\text{el}})$ do not vary within N time blocks [21], [25]. Then $\mathbf{H}[n]$ can be rewritten into a more concise form:

$$\mathbf{H}[n] = \sum_{k=1}^K \alpha_k e^{j2\pi f_d^k n T_s} \mathbf{a}_R(\theta_k^{\text{el}}, \theta_k^{\text{az}}) \mathbf{a}_T^H(\phi_k^{\text{az}}, \phi_k^{\text{el}}). \quad (10)$$

Define $\mathbf{G} = \mathbf{G}_d \text{diag}(\boldsymbol{\alpha})$, where $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_K] \in \mathbb{C}^{1 \times K}$ is the complex path gain vector and $\mathbf{G}_d = [\mathbf{g}_1^{\text{dop}}, \dots, \mathbf{g}_K^{\text{dop}}] \in \mathbb{C}^{N \times K}$ denotes the Doppler shift matrix with $\mathbf{g}_k^{\text{dop}} = [e^{j2\pi f_d^k T_s}, \dots, e^{j2\pi N f_d^k T_s}]^T \in \mathbb{C}^{N \times 1}$. Then (10) can be further

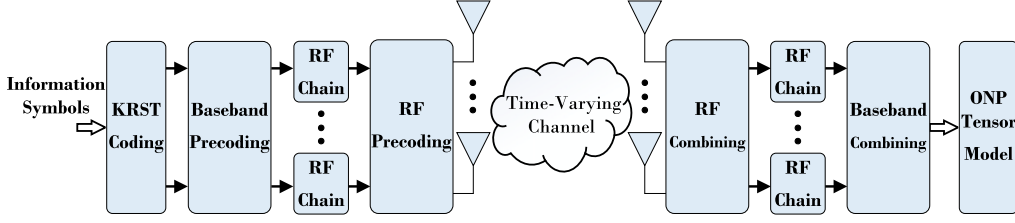


Fig. 1. Block-diagram of the ONP tensor model for the time-varying mmWave massive MIMO system.

expressed as

$$\mathbf{H}[n] = \mathbf{A}_R D_n \{\mathbf{G}\} \mathbf{A}_T^H \in \mathbb{C}^{N_R \times M_T}. \quad (11)$$

Note that it is reasonable to suppose that $\theta_{k_1}^{az} \neq \theta_{k_2}^{az}$, $\theta_{k_1}^{el} \neq \theta_{k_2}^{el}$, $\phi_{k_1}^{az} \neq \phi_{k_2}^{az}$, $\phi_{k_1}^{el} \neq \phi_{k_2}^{el}$ and $f_d^{k_1} \neq f_d^{k_2}$ for $k_1 \neq k_2$ and $k_1, k_2 = 1, \dots, K$ [7].

B. Nested PARAFAC Signal Model

Fig. 1 depicts the block-diagram of the proposed ONP tensor signal model for the time-varying hybrid mmWave massive MIMO system. Define $\mathbf{S} \in \mathbb{C}^{T \times R}$ as the aggregate information matrix to be sent over a time block, where R denotes the number of data streams and each time block contains T time slots. At the t -th time slot of the n -th block, the signal $\mathbf{S}$ is first encoded by the KRST coding matrix $\mathbf{C} \in \mathbb{C}^{P \times R}$, where P is the coding length. Then BS exploits a hybrid precoder $\mathbf{F}$, which is composed of a baseband precoding matrix $\mathbf{F}^{BB} \in \mathbb{C}^{M_{RF} \times R}$ and an RF precoding matrix $\mathbf{F}^{RF} \in \mathbb{C}^{M_T \times M_{RF}}$, with $\mathbf{F} = \mathbf{F}^{RF} \mathbf{F}^{BB}$. After transmitting via time-varying channels, the signal received by MS is

$$\mathbf{R}_{n,t} = \mathbf{A}_R D_n \{\mathbf{G}\} \mathbf{A}_T^H \mathbf{F} D_t \{\mathbf{S}\} \mathbf{C}^T + \mathbf{B}_{n,t}, \quad (12)$$

where $\mathbf{B}_{n,t} \in \mathbb{C}^{N_R \times P}$ represents the complex Gaussian noise. The signal $\mathbf{R}_{n,t}$ is then processed by a hybrid combiner $\mathbf{W} = \mathbf{W}^{RF} \mathbf{W}^{BB}$ at the MS, in which $\mathbf{W}^{RF} \in \mathbb{C}^{N_R \times N_{RF}}$ and $\mathbf{W}^{BB} \in \mathbb{C}^{N_{RF} \times N_{RF}}$ are RF and baseband combining matrices, respectively. The overall received signal at the t -th time slot of the n -th block is

$$\begin{aligned} \mathbf{Y}_{n,t}^{MS} &= \mathbf{W}^H \mathbf{R}_{n,t} \\ &= \mathbf{W}^H \mathbf{A}_R D_n \{\mathbf{G}\} \mathbf{A}_T^H \mathbf{F} D_t \{\mathbf{S}\} \mathbf{C}^T + \mathbf{W}^H \mathbf{B}_{n,t}. \end{aligned} \quad (13)$$

By defining $\mathbf{Q} = \mathbf{W}^H \mathbf{A}_R \in \mathbb{C}^{N_{RF} \times K}$, $\mathbf{P} = \mathbf{F}^T \mathbf{A}_T^* \in \mathbb{C}^{R \times K}$ and $\mathbf{N}_{n,t} = \mathbf{W}^H \mathbf{B}_{n,t}$, the formula (13) can be rewritten as

$$\mathbf{Y}_{n,t}^{MS} = \mathbf{Q} D_n \{\mathbf{G}\} \mathbf{P}^T D_t \{\mathbf{S}\} \mathbf{C}^T + \mathbf{N}_{n,t} \in \mathbb{C}^{N_{RF} \times P}. \quad (14)$$

Note that the KRST coding matrix $\mathbf{C}$, hybrid combining matrix $\mathbf{W}$ and hybrid precoding matrix $\mathbf{F}$ are known but the CSI is not known to neither BS nor MS.

By gathering $\mathbf{Y}_{n,t}^{MS}$ for all N blocks, the received signal can be expressed by a fourth-order nested PARAFAC tensor model $\mathcal{Y}^{MS}$ with size $T \times P \times N \times N_{RF}$. The associated noise tensor is denoted by $\mathcal{N}$. According to the definition of the nested PARAFAC model first introduced in [10], $\mathcal{Y}^{MS}$ can be further perceived as a nested PARAFAC model, which contains the following third-order outer PARAFAC model $\mathcal{Y} \in \mathbb{C}^{T \times P \times N_{RF} \times N}$

and inner PARAFAC model $\mathcal{X} \in \mathbb{C}^{N \times N_{RF} \times R}$ if the noise term is ignored, i.e.,

$$\mathcal{Y} = \mathcal{I}_{3,R} \times_1 \mathbf{S} \times_2 \mathbf{C} \times_3 [\mathcal{X}]_3, \quad (15)$$

$$\mathcal{X} = \mathcal{I}_{3,K} \times_1 \mathbf{G} \times_2 \mathbf{Q} \times_3 \mathbf{P}, \quad (16)$$

where $(\mathbf{S}, \mathbf{C}, [\mathcal{X}]_3)$ and $(\mathbf{G}, \mathbf{Q}, \mathbf{P})$ are their matrix factors, respectively. Moreover, $[\mathcal{X}]_3 = (\mathbf{G} \odot \mathbf{Q}) \mathbf{P}^T$ corresponds to mode-3 unfolding form of $\mathcal{X}$ with $x_{(n-1)N_{RF}+n_{RF},r}^{(3)} = x_{n,n_{RF},r}$, in which $x_{(n-1)N_{RF}+n_{RF},r}^{(3)}$ and $x_{n,n_{RF},r}$ respectively denote elements in $[\mathcal{X}]_3$ and $\mathcal{X}$ for $n = 1, \dots, N$, $n_{RF} = 1, \dots, N_{RF}$ and $r = 1, \dots, R$.

III. THE PROPOSED ONP-ITS-ALS ALGORITHM

The proposed ONP-ITS-ALS algorithm includes SVP-BALS algorithm and OR-TALS algorithm, which are used to fit the outer and the inner PARAFAC models, respectively.

A. SVP-BALS Algorithm From the Outer PARAFAC Model

The number of scatter paths K is usually unknown and needs to be estimated in advance. The minimum description length (MDL) [26]–[28] is able to detect the rank of matrices or tensors. If the noise term is ignored, then the received signal at the t -th time slot, i.e., $\mathcal{Y}^{MS}(t, :, :, :) \in \mathbb{C}^{P \times N \times N_{RF}}$, can be represented as

$$\mathcal{Y}^{MS}(t, :, :, :) = \mathcal{I}_{3,K} \times_1 \mathbf{G} \times_2 \mathbf{Q} \times_3 (\mathbf{C} D_t \{\mathbf{S}\} \mathbf{P}). \quad (17)$$

Clearly, the rank of tensor $\mathcal{Y}^{MS}(t, :, :, :)$ is the number of scatter paths K . In this subsection, we adopt a sequential MDL method to estimate the value of K . For simplicity of expression, we define $I_1 = N$, $I_2 = N_{RF}$, $I_3 = P$ and $I_T = N N_{RF} P$. In addition, the transpose of the mode- j unfolding of $\mathcal{Y}^{MS}(t, :, :, :)$ is denoted by $[\mathcal{G}]_j$, i.e., $[\mathcal{G}]_j = [\mathcal{Y}^{MS}(t, :, :, :)]_j^T$, where $j = 1, 2, 3$. By performing eigenvalue decomposition on the sample covariance matrix $\frac{I_j}{I_T} [\mathcal{G}]_j [\mathcal{G}]_j^H$, we obtain a set of eigenvalues in descending order $\{\lambda_{1,j}, \dots, \lambda_{I_j,j}\}$. In the j -th sequential procedure, the following global eigenvalues are used to the MDL criterion for detection

$$\lambda_{i,j}^{(G)} = \lambda_{i,1} + \dots + \lambda_{i,j}, \quad i = 1, \dots, I_j. \quad (18)$$

The value of K is estimated by $\hat{K} = \min\{\hat{K}_j\}_{j=1}^3$, where

$$\hat{K}_j = \arg \min_{k_j} \text{MDL}(k_j), \quad (19)$$

$$\text{MDL}(k_j) = \frac{k_j}{2} (2I_j - k_j) \log \frac{I_T}{I_j} + p_k \left([\mathcal{G}]_j, \hat{\nu}_k \right), \quad (20)$$

$\hat{\nu}_k$ is the maximum likelihood estimate of the parameter vector of the k -th model ν_k , and $p_k([\mathcal{G}]_j, \hat{\nu}_k)$ is the corresponding log-likelihood function with the following form [27], [28]

$$p_k([\mathcal{G}]_j, \hat{\nu}_k) = \frac{I_T}{I_j} (I_j - k_j) \log \frac{\frac{1}{I_j - k_j} \sum_{i=k_j+1}^{I_j} \lambda_{i,j}^{(G)}}{\prod_{i=k_j+1}^{I_j} \lambda_{i,j}^{(G) \frac{1}{I_j - k_j}}}. \quad (21)$$

The signal $\mathbf{S}$ and compound matrix $[\mathcal{X}]_3$ can be then estimated via the PARAFAC model $\mathcal{Y}$ as

$$\min_{\mathbf{S}, [\mathcal{X}]_3} \left\| \mathcal{Y} - \mathcal{I}_{3,R} \times_1 \hat{\mathbf{S}} \times_2 \mathbf{C} \times_3 [\hat{\mathcal{X}}]_3 \right\|_F^2 \quad \text{s.t.} \quad \text{rank}(\mathbf{H}[n]) \leq K. \quad (22)$$

According to the properties of PARAFAC decomposition [29], $\mathcal{Y}$ has the following mode-1 and mode-3 matrix representation

$$[\mathcal{Y}]_1 = (\mathbf{C} \odot [\mathcal{X}]_3) \mathbf{S}^T \in \mathbb{C}^{N_{\text{RF}}NP \times T}, \quad (23)$$

$$[\mathcal{Y}]_3 = (\mathbf{S} \odot \mathbf{C}) [\mathcal{X}]_3^T \in \mathbb{C}^{PT \times N_{\text{RF}}N}, \quad (24)$$

where $[\mathcal{Y}]_1 = [y_{l_1, l_2}^{(1)}]_{l_1, l_2=1}^{N_{\text{RF}}NP, T}$ and $[\mathcal{Y}]_3 = [y_{l_1, l_2}^{(3)}]_{l_1, l_2=1}^{PT, N_{\text{RF}}N}$ with $y_{(p-1)N_{\text{RF}}N+l, t}^{(1)} = y_{(t-1)P+p, l}^{(3)} = y_{t, p, l}$ for $p = 1, \dots, P$, $l = 1, \dots, N_{\text{RF}}N$ and $t = 1, \dots, T$.

The proposed SVP-BALS algorithm begins with a random initial value $\hat{\mathbf{S}}$. From (24), the least square (LS) solution of $[\mathcal{X}]_3$ can be obtained by

$$[\hat{\mathcal{X}}]_3 = \arg \min_{[\mathcal{X}]_3} \left\| [\mathcal{Y}]_3 - (\mathbf{S} \odot \mathbf{C}) [\mathcal{X}]_3^T \right\|_F^2. \quad (25)$$

The SVP-based algorithms [30] can effectively assist in estimating low-rank channels. By using the low-rank property of the massive MIMO channel matrix, the work [31] proposes a SVP-based algorithm combining Newton and gradient decent methods for joint CSI recovery. In point-to-point mmWave massive MIMO systems, a large number of antennas are configured at BS and MS. Since the number of scatter paths K is generally small, $\mathbf{H}[n]$ in (11) for all $n = 1, \dots, N$ are low-rank matrices [19]–[21]. It is reasonable to assume that the number of RF chains is greater than that of scatter paths in this paper. In addition, the entries in hybrid combining and precoding matrices $\mathbf{W}$ and $\mathbf{F}$ have random properties (The detail structures of $\mathbf{W}$ and $\mathbf{F}$ are given in Section III.B). Hence $[\hat{\mathcal{X}}]_3^n = [\hat{\mathcal{X}}]_3((n-1)N_{\text{RF}}+1 : (n-1)N_{\text{RF}}, :)$ are low-rank matrices for all $n = 1, \dots, N$ and $[\hat{\mathcal{X}}]_3$ can be optimized at every iteration via SVP, i.e.,

$$([\hat{\mathcal{X}}]_3^n)_{\text{new}} = \text{SVP}([\hat{\mathcal{X}}]_3^n) = \sum_{k=1}^K \sigma_k^n \mathbf{u}_k^n \mathbf{v}_k^n, \quad (26)$$

in which σ_k^n is the k -th largest singular value of $[\hat{\mathcal{X}}]_3^n$, while $\mathbf{u}_k^n$ and $\mathbf{v}_k^n$ represent the corresponding left and right singular vectors. Then the current estimated result $[\hat{\mathcal{X}}]_3$ is updated by

$$([\hat{\mathcal{X}}]_3)_{\text{new}} = \left[([\hat{\mathcal{X}}]_3^n)_{\text{new}}^T, \dots, ([\hat{\mathcal{X}}]_3^N)_{\text{new}}^T \right]^T. \quad (27)$$

As will be shown in Section V.B, formulas (26) and (27) cost less processing time but significantly improve the joint estimation performance. Next, $\mathbf{S}$ can be estimated with known $\mathbf{C}$ and newly

TABLE I
THE PROPOSED ONP-ITS-ALS ALGORITHM

First stage: SVP-BALS Algorithm

- Input:** Received signal tensor $\mathcal{Y}^{\text{MS}}$, KRST coding matrix $\mathbf{C}$, and error threshold $\varepsilon_{\text{th}} = 1 \times 10^{-4}$.
- 1) Obtain eigenvalues by performing eigenvalue decomposition on $\frac{1}{I_T} [\mathcal{G}]_j [\mathcal{G}]_j^H$ for all $j = 1, 2, 3$;
 - 2) Compute global eigenvalues by (18);
 - 3) Compute K_j by (19)–(21) for all $j = 1, 2, 3$;
 - 4) Estimate the number of scatter paths K by $\hat{K} = \min \left\{ \hat{K}_j \right\}_{j=1}^3$;
 - 5) Initialize $\hat{\mathbf{S}}^0$ and set $i = 0$;
 - 6) $i = i + 1$;
 - 7) Calculate the LS solution $([\hat{\mathcal{X}}]_3)^i$ by (25);
 - 8) For $n = 1, \dots, N$
 - a) Apply SVP to the low-rank matrix $([\hat{\mathcal{X}}]_3^n)^i$ according to (26);
 - b) Reconstruct $([\hat{\mathcal{X}}]_3)_{\text{new}}^i$ by (27);
 - 9) End for
 - 10) Calculate the LS solution $\hat{\mathbf{S}}^i$ by (28);
 - 11) Calculate $\varpi(i)$ by (29) in the i -th iteration;
 - 12) Repeat steps 6)–12) until $|\varpi(i) - \varpi(i-1)|/\varpi(i) \leq \varepsilon_{\text{th}}$;
 - 13) Eliminate scaling ambiguity using (30).
- Output:** The estimated signal matrix $\hat{\mathbf{S}}$ and compound matrix $[\hat{\mathcal{X}}]_3$.

obtained $([\hat{\mathcal{X}}]_3)_{\text{new}}$, i.e.,

$$\hat{\mathbf{S}} = \arg \min_{\mathbf{S}} \left\| [\mathcal{Y}]_1 - (\mathbf{C} \odot ([\hat{\mathcal{X}}]_3)_{\text{new}}) \mathbf{S}^T \right\|_F^2. \quad (28)$$

The error for the i -th iteration can be computed from

$$\varpi(i) = \left\| [\mathcal{Y}]_3 - (\hat{\mathbf{S}}^i \odot \mathbf{C}) ([\hat{\mathcal{X}}]_3^n)_{\text{new}}^i \right\|_F^2. \quad (29)$$

The algorithm is supposed to converge when $|\varpi(i) - \varpi(i-1)|/\varpi(i) \leq \varepsilon_{\text{th}}$, where ε_{th} denotes the error tolerance.

There are intrinsic ambiguities of PARAFAC tensor decomposition [10], [12], [29]. In order to obtain unambiguous estimations of $\mathbf{S}$ and $[\mathcal{X}]_3$, we set the first row of signal matrix $\mathbf{S}$ to all-ones elements as in [10], [12] and there are

$$\begin{aligned} \hat{\mathbf{S}} &\leftarrow \hat{\mathbf{S}} / \text{diag}(\hat{\mathbf{S}}_{1,:}), \\ [\hat{\mathcal{X}}]_3 &\leftarrow [\hat{\mathcal{X}}]_3 \text{diag}(\hat{\mathbf{S}}_{1,:}). \end{aligned} \quad (30)$$

The proposed SVP-BALS algorithm is summarized in Table I.

Discussion: By exploiting the matricizing operation, the mode-2 unfolding matrix of $\mathcal{Y}$ is given by

$$[\mathcal{Y}]_2 = ([\mathcal{X}]_3 \odot \mathbf{S}) \mathbf{C}^T \in \mathbb{C}^{N_{\text{RF}}NT \times P} \quad (31)$$

with $y_{(t-1)N_{\text{RF}}N+l, p}^{(2)} = y_{t, p, l}$ for $t = 1, \dots, T$, $l = 1, \dots, N_{\text{RF}}N$ and $p = 1, \dots, P$. Note that $\mathbf{C} \in \mathbb{C}^{P \times R}$ is chosen beforehand, such as an random matrix or the Vandermonde matrix. If $P \geq R$, then $\mathbf{C}$ has full column-rank. Actually, we can left multiply $[\mathcal{Y}]_2$ by the Moore-Penrose pseudo-inverse of $\mathbf{C}^T$ and further obtain $[\mathcal{X}]_3 \odot \mathbf{S} = [\mathcal{Y}]_2 (\mathbf{C}^T)^\dagger$. Then the singular value decomposition (SVD) based algorithm [32], [33] can be used to estimate $\mathbf{S}$ and $[\mathcal{X}]_3$ by matricizing each column of $[\mathcal{X}]_3 \odot \mathbf{S}$. Such a SVD-based algorithm provides a

relatively faster estimation speed than the proposed SVP-BALS algorithm, yet its disadvantages are also apparent. For example, the SVD-based algorithm is a purely algebraic closed-form solution and leads to a loss in the estimation performance [34]. The recovery of channel parameters is relevant in the estimation accuracy of signal $\mathbf{S}$ and compound matrix $[\mathcal{X}]_3$. Due to the influence of error accumulation, the channel estimation suffers from performance degradation, which will be shown in Section V.B and V.C. Besides, this SVD-based algorithm can not guarantee the unique determination in the case of $P < R$. As analyzed in Section IV.A, the proposed SVP-BALS scheme can work with more relax uniqueness conditions.

B. OR-TALS Algorithm From the Inner Real-Valued PARAFAC Model

The estimation of $[\mathcal{X}]_3$ is harvested by the SVP-BALS algorithm. In order to reduce the computational complexity and improve channel parameters estimation accuracy, we use the forward-backward and unitary transformation technique to convert the inner complex-valued PARAFAC model $\mathcal{X}$ into a real-valued one [35].

By reshaping $[\mathcal{X}]_3$, the following mode-1 unfolding matrix of $\mathcal{X}$ can be obtained, in the absence of noise, as

$$[\mathcal{X}]_1 = (\mathbf{Q} \odot \mathbf{P}) \mathbf{G}^T = (\mathbf{W}^H \mathbf{A}_R \odot \mathbf{F}^T \mathbf{A}_T^*) \mathbf{G}^T, \quad (32)$$

where $[\mathcal{X}]_1 = [x_{l,n}^{(1)}]_{l,n=1}^{R N_{\text{RF}} \times N}$ with $x_{(n_{\text{RF}}-1)R+r,n}^{(1)} = x_{n,n_{\text{RF}},r}$ for $n = 1, \dots, N$, $n_{\text{RF}} = 1, \dots, N_{\text{RF}}$ and $r = 1, \dots, R$.

We devise the hybrid combining and precoding matrices $\mathbf{W}$ and $\mathbf{F}$ as centro-Hermitian matrices, i.e.,

$$\begin{cases} \mathbf{W} = \mathbf{\Pi}_{N_R} \mathbf{W}^* \mathbf{\Pi}_{N_{\text{RF}}} \\ \mathbf{F} = \mathbf{\Pi}_{M_T} \mathbf{F}^* \mathbf{\Pi}_R \end{cases}. \quad (33)$$

In practice communication systems, both $\mathbf{F}^{\text{RF}}$ and $\mathbf{W}^{\text{RF}}$ are accomplished by analog phase shifters, which only change the phase of transmitted signals. Therefore, the constant modulus constraint is imposed on the elements of $\mathbf{F}^{\text{RF}}$ and $\mathbf{W}^{\text{RF}}$. The total transmission power constraint is enforced by normalizing $\mathbf{F}^{\text{BB}}$ such that $\|\mathbf{F}^{\text{RF}} \mathbf{F}^{\text{BB}}\|_F^2 = R$. The centro-Hermitian structure placed on $\mathbf{W}$ and $\mathbf{F}$ do not break these constraints. In simulations, $\mathbf{W}$ and $\mathbf{F}$ are constructed by (1.5.1)–(1.5.4) in [36], in which $\mathbf{A}$, $\mathbf{B}$, $\mathbf{a}$ and $\mathbf{b}$ are given by $\exp(-j2\pi u)/\sqrt{M_T}$ or $\exp(-j2\pi u)/\sqrt{N_R}$, $j = \sqrt{-1}$, and u is chosen from the independent and identically distributed (i.i.d.) standard uniform distribution. This design scheme can ensure the essential uniqueness of the constructed tensor model and helps to convert the inner complex-valued PARAFAC model $\mathcal{X}$ into a real-valued one. This design scheme is not optimal in general. However, we find that it is also a good choice, which has great signal detection and channel parameters estimation performance in simulations. The optimal design of hybrid combining and precoding matrices $\mathbf{W}$ and $\mathbf{F}$ is beyond the scope of this paper and will be explored in our future work.

For simplicity, we assume that the ULAs are set at BS and MS. Letting the center of antenna array be the phase reference [37],

the steering vector can be expressed as

$$\mathbf{a}_{\text{ULA}}(\psi) = [e^{j2\pi d(\frac{M-1}{2})\psi/\lambda}, e^{j2\pi d(\frac{M-1}{2}-1)\psi/\lambda}, \dots, e^{j2\pi d(-\frac{M-1}{2})\psi/\lambda}]^T. \quad (34)$$

It can be readily checked that $\mathbf{a}_T(\phi_k)$ and $\mathbf{a}_R(\theta_k)$ are conjugate centrosymmetric, and their corresponding steering matrices are left Π -real [38], i.e.,

$$\begin{cases} \mathbf{\Pi}_{M_T} \mathbf{A}_T^* = \mathbf{A}_T \\ \mathbf{\Pi}_{N_R} \mathbf{A}_R^* = \mathbf{A}_R \end{cases}. \quad (35)$$

In order to convert $\mathcal{X}$ into a real-valued PARAFAC model, a feasible scheme is to construct a centro-Hermitian matrix $\bar{\mathbf{Z}} = [[\mathcal{X}]_1, \mathbf{\Pi}_{R N_{\text{RF}}} [\mathcal{X}]_1^* \mathbf{\Pi}_N]$. Firstly, we rewrite $\mathbf{\Pi}_{R N_{\text{RF}}} [\mathcal{X}]_1^* \mathbf{\Pi}_N$ as

$$\begin{aligned} \mathbf{\Pi}_{R N_{\text{RF}}} [\mathcal{X}]_1^* \mathbf{\Pi}_N &\stackrel{(a)}{=} (\mathbf{\Pi}_{N_{\text{RF}}} \otimes \mathbf{\Pi}_R) (\mathbf{W}^T \mathbf{A}_R^* \odot \mathbf{F}^H \mathbf{A}_T) \mathbf{G}^H \mathbf{\Pi}_N \\ &\stackrel{(b)}{=} (\mathbf{\Pi}_{N_{\text{RF}}} \mathbf{W}^T \mathbf{\Pi}_{N_R} \mathbf{A}_R \odot \mathbf{\Pi}_R \mathbf{F}^H \mathbf{\Pi}_{M_T} \mathbf{A}_T^*) \mathbf{G}^H \mathbf{\Pi}_N \\ &\stackrel{(c)}{=} (\mathbf{W}^H \mathbf{A}_R \odot \mathbf{F}^T \mathbf{A}_T^*) \mathbf{G}^H \mathbf{\Pi}_N \\ &= (\mathbf{Q} \odot \mathbf{P}) \mathbf{G}^H \mathbf{\Pi}_N, \end{aligned} \quad (36)$$

where (a) follows from $\mathbf{\Pi}_{R N_{\text{RF}}} = \mathbf{\Pi}_{N_{\text{RF}}} \otimes \mathbf{\Pi}_R$ and (35), (b) follows from the property (1), and (c) follows from the formula (33). Then according to formulas (32) and (36), $\bar{\mathbf{Z}}$ can be rewritten as

$$\bar{\mathbf{Z}} = (\mathbf{Q} \odot \mathbf{P}) [\mathbf{G}^T, \mathbf{G}^H \mathbf{\Pi}_N]. \quad (37)$$

By introducing the unitary matrices $\mathbf{\Gamma}_{N_R}$, $\mathbf{\Gamma}_{M_T}$ and $\mathbf{\Gamma}_{2N}$, $\bar{\mathbf{Z}}$ is converted into the real-valued PARAFAC model $\mathcal{Z} \in \mathbb{R}^{N_{\text{RF}} \times R \times 2N}$ [36], [39] with following mode-3 matrix representation

$$\begin{aligned} [\mathcal{Z}]_3 &= (\mathbf{\Gamma}_{N_{\text{RF}}}^H \otimes \mathbf{\Gamma}_R^H) \bar{\mathbf{Z}} \mathbf{\Gamma}_{2N} \in \mathbb{R}^{R N_{\text{RF}} \times 2N} \\ &= ((\mathbf{\Gamma}_{N_{\text{RF}}}^H \mathbf{Q}) \odot (\mathbf{\Gamma}_R^H \mathbf{P})) [\mathbf{G}^T, \mathbf{G}^H \mathbf{\Pi}_N] \mathbf{\Gamma}_{2N} \\ &= (\bar{\mathbf{A}}_R \odot \bar{\mathbf{A}}_T) \bar{\mathbf{G}}^T, \end{aligned} \quad (38)$$

where $\bar{\mathbf{A}}_R = \mathbf{\Gamma}_{N_{\text{RF}}}^H \mathbf{Q}$, $\bar{\mathbf{A}}_T = \mathbf{\Gamma}_R^H \mathbf{P}$ and $\bar{\mathbf{G}}^T = [\mathbf{G}^T, \mathbf{G}^H \mathbf{\Pi}_N] \mathbf{\Gamma}_{2N}$ are real-valued matrix factors. Moreover, $\mathbf{\Gamma}_{N_R}$, $\mathbf{\Gamma}_{M_T}$ and $\mathbf{\Gamma}_{2N}$ enjoy the following forms with even or odd order

$$\begin{aligned} \mathbf{\Gamma}_{2m} &= \frac{1}{\sqrt{2}} \begin{bmatrix} \mathbf{I}_m & j\mathbf{I}_m \\ \mathbf{\Pi}_m & -j\mathbf{\Pi}_m \end{bmatrix}, \\ \mathbf{\Gamma}_{2m+1} &= \frac{1}{\sqrt{2}} \begin{bmatrix} \mathbf{I}_m & \mathbf{0} & j\mathbf{I}_m \\ \mathbf{0}^T & \sqrt{2} & \mathbf{0}^T \\ \mathbf{\Pi}_m & \mathbf{0} & -j\mathbf{\Pi}_m \end{bmatrix}. \end{aligned} \quad (39)$$

We then reshape $[\mathcal{Z}]_3$ into the following two compact forms

$$[\mathcal{Z}]_1 = (\bar{\mathbf{A}}_T \odot \bar{\mathbf{G}}) \bar{\mathbf{A}}_R^T \in \mathbb{R}^{2N R \times N_{\text{RF}}}, \quad (40)$$

$$[\mathcal{Z}]_2 = (\bar{\mathbf{G}} \odot \bar{\mathbf{A}}_R) \bar{\mathbf{A}}_T^T \in \mathbb{R}^{2N_{\text{RF}} N \times R}. \quad (41)$$

According to the existing TALS algorithm [7], [11], the three matrix factors $\bar{\mathbf{A}}_R$, $\bar{\mathbf{A}}_T$ and $\bar{\mathbf{G}}$ can be estimated iteratively by using (38), (40) and (41).

However, since all three factor matrices need to be estimated, the TALS algorithm requires more iterations to achieve convergence, and would easily fall into the local loop when factors are almost collinear [40], [41]. In particular, the TALS algorithm has

high complexity for complex-valued PARAFAC models. Since the inner PARAFAC model has been converted into a real one, to accelerate the convergence, the OR line search process is performed before TALS to predict the factor matrices used for the i -th iteration, i.e., the OR-TALS algorithm.

Define

$$\begin{cases} \bar{\mathbf{A}}_{\mathbf{R}}^{\text{new}} = \hat{\mathbf{A}}_{\mathbf{R}}^{i-2} + \rho \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i \\ \bar{\mathbf{A}}_{\mathbf{T}}^{\text{new}} = \hat{\mathbf{A}}_{\mathbf{T}}^{i-2} + \rho \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i \\ \bar{\mathbf{G}}^{\text{new}} = \hat{\mathbf{G}}^{i-2} + \rho \mathbf{M}_{\bar{\mathbf{G}}}^i \end{cases}, \quad (42)$$

where $\mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i = \hat{\mathbf{A}}_{\mathbf{R}}^{i-1} - \hat{\mathbf{A}}_{\mathbf{R}}^{i-2}$, $\mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i = \hat{\mathbf{A}}_{\mathbf{T}}^{i-1} - \hat{\mathbf{A}}_{\mathbf{T}}^{i-2}$ and $\mathbf{M}_{\bar{\mathbf{G}}}^i = \hat{\mathbf{G}}^{i-1} - \hat{\mathbf{G}}^{i-2}$ are the search directions in the i -th iteration, and ρ is the corresponding relaxation factor. Given the estimates of $\bar{\mathbf{A}}_{\mathbf{R}}$, $\bar{\mathbf{A}}_{\mathbf{T}}$ and $\bar{\mathbf{G}}$ in the $(i-2)$ -th and $(i-1)$ -th iterations, the error in the i -th iteration is expressed as

$$\begin{aligned} \varsigma(i) &= \left\| [\mathcal{Z}]_3 - (\bar{\mathbf{A}}_{\mathbf{R}}^{\text{new}} \odot \bar{\mathbf{A}}_{\mathbf{T}}^{\text{new}}) \bar{\mathbf{G}}^{\text{newT}} \right\|_{\text{F}}^2 \\ &= \left\| ((\hat{\mathbf{A}}_{\mathbf{R}}^{i-2} + \rho \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i) \odot (\hat{\mathbf{A}}_{\mathbf{T}}^{i-2} + \rho \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i)) \right. \\ &\quad \left. (\hat{\mathbf{G}}^{i-2} + \rho \mathbf{M}_{\bar{\mathbf{G}}}^i)^{\text{T}} - [\mathcal{Z}]_3 \right\|_{\text{F}}^2. \end{aligned} \quad (43)$$

The goal is to compute the optimal ρ for the minimization of $\varsigma(i)$ and speed up the convergence. Define matrices $\mathbf{T}_3$, $\mathbf{T}_2$, $\mathbf{T}_1$ and $\mathbf{T}_0$ as

$$\begin{cases} \mathbf{T}_3 = (\mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i \odot \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i) (\mathbf{M}_{\bar{\mathbf{G}}}^i)^{\text{T}} \\ \mathbf{T}_2 = \left(\hat{\mathbf{A}}_{\mathbf{R}}^{i-2} \odot \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i + \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i \odot \hat{\mathbf{A}}_{\mathbf{T}}^{i-2} \right) (\mathbf{M}_{\bar{\mathbf{G}}}^i)^{\text{T}} + \\ \quad (\mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i \odot \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i) (\hat{\mathbf{G}}^{i-2})^{\text{T}} \\ \mathbf{T}_1 = \left(\hat{\mathbf{A}}_{\mathbf{R}}^{i-2} \odot \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{T}}}^i + \mathbf{M}_{\bar{\mathbf{A}}_{\mathbf{R}}}^i \odot \hat{\mathbf{A}}_{\mathbf{T}}^{i-2} \right) (\hat{\mathbf{G}}^{i-2})^{\text{T}} + \\ \quad \left(\hat{\mathbf{A}}_{\mathbf{R}}^{i-2} \odot \hat{\mathbf{A}}_{\mathbf{T}}^{i-2} \right) (\mathbf{M}_{\bar{\mathbf{G}}}^i)^{\text{T}} \\ \mathbf{T}_0 = \left(\hat{\mathbf{A}}_{\mathbf{R}}^{i-2} \odot \hat{\mathbf{A}}_{\mathbf{T}}^{i-2} \right) (\hat{\mathbf{G}}^{i-2})^{\text{T}} - [\mathcal{Z}]_3 \end{cases}. \quad (44)$$

With the definition (44), $\varsigma(i)$ can be rewritten as

$$\begin{aligned} \varsigma(i) &= \left\| \rho^3 \mathbf{T}_3 + \rho^2 \mathbf{T}_2 + \rho \mathbf{T}_1 + \mathbf{T}_0 \right\|_{\text{F}}^2 \\ &= \left\| \mathbf{T} \mathbf{u} \right\|_{\text{F}}^2 \\ &= \mathbf{u}^{\text{H}} \mathbf{T}^{\text{H}} \mathbf{T} \mathbf{u}, \end{aligned} \quad (45)$$

where $\mathbf{T} = [\text{vec}(\mathbf{T}_3), \text{vec}(\mathbf{T}_2), \text{vec}(\mathbf{T}_1), \text{vec}(\mathbf{T}_0)] \in \mathbb{R}^{2NRN_{\text{RF}} \times 4}$, and $\mathbf{u} = [\rho^3, \rho^2, \rho, 1]^{\text{T}}$. Since $\bar{\mathbf{A}}_{\mathbf{R}}$, $\bar{\mathbf{A}}_{\mathbf{T}}$ and $\bar{\mathbf{G}}$ are real-valued matrices, we know $\varsigma(i) = \mathbf{u}^{\text{T}} \mathbf{T}^{\text{T}} \mathbf{T} \mathbf{u} = \mathbf{u}^{\text{T}} \Delta \mathbf{u}$, where $\Delta = \mathbf{T}^{\text{T}} \mathbf{T} \in \mathbb{R}^{4 \times 4}$. Moreover, $\varsigma(i)$ is a polynomial of degree six in the variable ρ :

$$\begin{aligned} \varsigma(i) &= \Delta_{1,1} \rho^6 + 2\Delta_{2,1} \rho^5 + (\Delta_{2,2} + 2\Delta_{3,1}) \rho^4 + \\ &\quad 2(\Delta_{4,1} + \Delta_{3,2}) \rho^3 + (\Delta_{3,3} + 2\Delta_{4,2}) \rho^2 + \\ &\quad 2\Delta_{4,3} \rho + \Delta_{4,4}. \end{aligned} \quad (46)$$

To find the optimal ρ , we compute

$$\begin{aligned} \frac{d\varsigma(i)}{d\rho} &= 6\Delta_{1,1} \rho^5 + 10\Delta_{2,1} \rho^4 + 4(\Delta_{2,2} + 2\Delta_{3,1}) \rho^3 + \\ &\quad 6(\Delta_{4,1} + \Delta_{3,2}) \rho^2 + 2(\Delta_{3,3} + 2\Delta_{4,2}) \rho + 2\Delta_{4,3} \\ &= 0. \end{aligned} \quad (47)$$

It provides five possible roots, and we choose the one that gives the smallest $\varsigma(i)$ as the optimal value of ρ .

Compared with the traditional TALS algorithm based on the complex-valued PARAFAC model, the proposed algorithm can greatly reduce the number of iterations and improve the convergence speed while without losing fitting accuracy. Its performance will be further verified and analyzed in Section V.

Afterwards, we aim at how to recover the channel parameters $(\hat{\theta}_k, \hat{\phi}_k, \hat{f}_d^k, \hat{\alpha}_k)$ from the estimated real-valued matrix factors $(\hat{\mathbf{A}}_{\mathbf{R}}, \hat{\mathbf{A}}_{\mathbf{T}}, \hat{\mathbf{G}})$. The estimations of $\mathbf{Q}$ and $\mathbf{P}$ are retrieved by $\hat{\mathbf{Q}} = \Gamma_{N_{\text{RF}}} \hat{\mathbf{A}}_{\mathbf{R}}$ and $\hat{\mathbf{P}} = \Gamma_R \hat{\mathbf{A}}_{\mathbf{T}}$, respectively. Since each column of $\hat{\mathbf{Q}}$ and $\hat{\mathbf{P}}$ are connected to $\hat{\theta}_k$ and $\hat{\phi}_k$, respectively, we extract AoAs and AoDs as follows:

$$\hat{\theta}_k = \arg \max_{\theta_k} \frac{|\hat{\mathbf{q}}_k^{\text{H}} \mathbf{a}_{\mathbf{R}}(\theta_k)|}{\|\hat{\mathbf{q}}_k\|_2 \|\mathbf{a}_{\mathbf{R}}(\theta_k)\|_2}, \quad (48)$$

$$\hat{\phi}_k = \arg \max_{\phi_k} \frac{|\hat{\mathbf{p}}_k^{\text{H}} \mathbf{a}_{\mathbf{T}}(\phi_k)|}{\|\hat{\mathbf{p}}_k\|_2 \|\mathbf{a}_{\mathbf{T}}(\phi_k)\|_2}, \quad (49)$$

where $\hat{\mathbf{q}}_k$ and $\hat{\mathbf{p}}_k$ stand for the k -th column of $\hat{\mathbf{Q}}$ and $\hat{\mathbf{P}}$, respectively. The estimations of θ_k and ϕ_k for all $k = 1, \dots, K$ can be obtained by the one-dimensional searching method. After that, we restore $\hat{\mathbf{Q}}$ and $\hat{\mathbf{P}}$, and then obtain $\hat{\mathbf{G}}$ from $\hat{\mathbf{G}} = [(\hat{\mathbf{Q}} \odot \hat{\mathbf{P}})^{\dagger} [\mathcal{X}]_1]^{\text{T}}$. In fact, each column of $\hat{\mathbf{G}}$ can be expressed as $\alpha_k \mathbf{g}_k^{\text{dop}}$. Similar to extracting $\hat{\theta}_k$ and $\hat{\phi}_k$, the Doppler shift can be retrieved from

$$\hat{f}_d^k = \arg \max_{f_d^k} \frac{|\hat{\mathbf{g}}_k^{\text{H}} \mathbf{g}_k^{\text{dop}}(f_d^k)|}{\|\hat{\mathbf{g}}_k\|_2 \|\mathbf{g}_k^{\text{dop}}(f_d^k)\|_2}, \quad (50)$$

where $\hat{\mathbf{g}}_k$ stands for the k -th column of $\hat{\mathbf{G}}$, and f_d^k encompasses the sector of interests for one-dimensional search. The estimated Doppler shift matrix $\hat{\mathbf{G}}_{\text{d}}$ is then reconstructed, and the complex path gains are estimated via

$$\hat{\alpha}_k = \frac{1}{N} \sum_{n=1}^N \hat{\mathbf{G}}(n, k) / \hat{\mathbf{G}}_{\text{d}}(n, k). \quad (51)$$

Now that all channel parameters $(\hat{\theta}_k, \hat{\phi}_k, \hat{f}_d^k, \hat{\alpha}_k)$ for $k = 1, \dots, K$ have been estimated, we can rebuild $\hat{\mathbf{H}}[n]$ through (11). The estimation procedures of the proposed OR-TALS algorithm are shown in Table II.

According to the properties of PARAFAC decomposition [29], the estimated matrix factors $(\hat{\mathbf{A}}_{\mathbf{R}}, \hat{\mathbf{A}}_{\mathbf{T}}, \hat{\mathbf{G}})$ inevitably have unknown scaling and permutation ambiguities compared with their true values. However, the recovery of $\hat{\theta}_k$ and $\hat{\phi}_k$ by using (48) and (49) involves one-dimensional search, which is performed separately for each column of $\hat{\mathbf{Q}}$ or $\hat{\mathbf{P}}$. In addition, all

TABLE II
THE PROPOSED ONP-ITS-ALS ALGORITHM

Second stage: OR-TALS Algorithm
Input: The estimated compound matrix $[\mathcal{X}]_3$, hybrid combining matrix $\mathbf{W}$, hybrid precoding matrix $\mathbf{F}$, and error threshold $\varepsilon_{\text{th}} = 1 \times 10^{-4}$.
1) Convert $\mathcal{X}$ into the real-valued PARAFAC model $\mathcal{Z}$ with mode-1, mode-2 and mode-3 unfolding matrices $[\mathcal{Z}]_1$, $[\mathcal{Z}]_2$ and $[\mathcal{Z}]_3$; 2) Initialize $\hat{\mathbf{A}}_R^0$, $\hat{\mathbf{A}}_T^0$, $\hat{\mathbf{A}}_R^1$, $\hat{\mathbf{A}}_T^1$, $\hat{\mathbf{G}}^0$, $\hat{\mathbf{G}}^1$, and set $i = 1$, $\rho = 1$; 3) $i = i + 1$; 4) Calculate $\bar{\mathbf{A}}_R^{\text{new}}$, $\bar{\mathbf{A}}_T^{\text{new}}$ and $\bar{\mathbf{G}}^{\text{new}}$ by (42); 5) Find the optimal ρ according to (43)-(47); 6) Rebuild $\bar{\mathbf{A}}_R^{\text{new}}$ and $\bar{\mathbf{A}}_T^{\text{new}}$ by substituting ρ into (42); 7) Calculate the LS solution $\hat{\mathbf{G}}^i$ via using $\bar{\mathbf{A}}_R^{\text{new}}$, $\bar{\mathbf{A}}_T^{\text{new}}$ and $[\mathcal{Z}]_3$; 8) Calculate the LS solution $\hat{\mathbf{A}}_R^i$ via using $\bar{\mathbf{A}}_T^{\text{new}}$, $\hat{\mathbf{G}}^i$ and $[\mathcal{Z}]_1$; 9) Calculate the LS solution $\hat{\mathbf{A}}_T^i$ via using $\hat{\mathbf{G}}^i$, $\bar{\mathbf{A}}_R^{\text{new}}$ and $[\mathcal{Z}]_2$; 10) Calculate $\varsigma(i)$; 11) Repeat steps 3)-11) until $ \varsigma(i) - \varsigma(i-1) /\varsigma(i) \leq \varepsilon_{\text{th}}$; 12) Recover $\hat{\mathbf{Q}} = \mathbf{\Gamma}_{N_{\text{RF}}} \hat{\mathbf{A}}_R$ and $\hat{\mathbf{P}} = \mathbf{\Gamma}_R \hat{\mathbf{A}}_T$; 13) Extract $\hat{\theta}_k$ and $\hat{\phi}_k$ for $k = 1, \dots, K$ by (48) and (49); 14) Restore $\hat{\mathbf{Q}}$ and $\hat{\mathbf{P}}$ and calculate $\hat{\mathbf{G}} = \left[(\hat{\mathbf{Q}} \odot \hat{\mathbf{P}})^\dagger [\mathcal{X}]_1 \right]^T$; 15) Extract $\hat{f}_d^k$ for $k = 1, \dots, K$ by (50); 16) Restore $\hat{\mathbf{G}}_d$; 17) Estimate $\hat{\alpha}_k$ for $k = 1, \dots, K$ by (51); 18) Rebuild the time-varying mmWave channel $\hat{\mathbf{H}}[n]$ by (11).
Output: Estimated channel parameters $(\hat{\theta}_k, \hat{\phi}_k, \hat{f}_d^k, \hat{\alpha}_k)$ for $k = 1, \dots, K$ and time-varying mmWave channel $\hat{\mathbf{H}}[n]$ for $n = 1, \dots, N$.

columns of $\hat{\mathbf{Q}}$ or $\hat{\mathbf{P}}$ share the same scaling ambiguity. Therefore, $\hat{\theta}_k$ and $\hat{\phi}_k$, as well as $\hat{f}_d^k$ and $\hat{\alpha}_k$, do not be confined to the scaling ambiguity [7], [23]. Furthermore, the permutation ambiguity can be neglected since all factor matrices share the same permutation matrix. As a result, it is reasonable to ignore the unknown scaling and permutation ambiguities for the proposed OR-TALS algorithm.

Note that the proposed OR-TALS algorithm is not subject to the conjugate centrosymmetric property of the steering vectors in (34). The real-valued PARAFAC model can also be deduced in typical ULAs and UPAs, and the associated derivation processes are shown in Appendix in detail.

IV. UNIQUENESS AND COMPLEXITY

In this section, we delve into the essential uniqueness conditions required for the proposed algorithm. This is related to whether the accurate information symbols and channel parameters can be recovered from the estimated factor matrices. The complexity of the proposed algorithm is also discussed in this section.

A. Uniqueness Issues

We commence with the essential uniqueness condition of $\mathcal{Y}$. The k -rank of a matrix $\mathbf{A}$ represents the maximum number of linearly independent columns in $\mathbf{A}$. It should be noted that k -rank is less than or equal to the rank of the conventional matrix. In other words, if $\mathbf{A}$ is full column-rank, then it is also full

k -rank. According to the Kruskal's condition [42], if

$$k(\mathbf{S}) + k(\mathbf{C}) + k([\mathcal{X}]_3) \geq 2R + 2, \quad (52)$$

then the triple $(\mathbf{S}, \mathbf{C}, [\mathcal{X}]_3)$ is unique up to scaling and permutation ambiguities. The KRST coding matrix $\mathbf{C}$ is known to the MS. By applying the Lemma 1 in [29], we know that if $\mathbf{S}$ (or $[\mathcal{X}]_3$) and $\mathbf{C}$ are full column-rank matrices, then $[\mathcal{X}]_3$ and $\mathbf{S}$ are only unique to a diagonal scaling matrix, i.e.,

$$[\bar{\mathcal{X}}]_3 = [\mathcal{X}]_3 \mathbf{\Lambda}, \quad \bar{\mathbf{S}} = \mathbf{S} \mathbf{\Lambda}^{-1}, \quad (53)$$

where $[\bar{\mathcal{X}}]_3$ and $\bar{\mathbf{S}}$ are alternative solutions of $[\mathcal{X}]_3$ and $\mathbf{S}$, $\mathbf{\Lambda}$ is a diagonal scaling matrix. The signal matrix $\mathbf{S} \in \mathbb{C}^{T \times R}$ is randomly generated and modulated by 64QAM. It is full column-rank, i.e., $k(\mathbf{S}) = R$, with very high probability when T is large enough. Therefore, the proposed SVP-BALS algorithm can uniquely estimate signal $\mathbf{S}$ and compound matrix $[\mathcal{X}]_3$ even in the single path case. But the inequality (52) can not be established when $K = 1$. If T is small, $\mathbf{S}$ is likely to be rank deficient due to the finite symbol constellation. However, simulation results show that the proposed algorithm can still ensure uniqueness with small T . This is because $\mathbf{C}$ is known, there is a more relaxed uniqueness condition for the constructed PARAFAC model $\mathcal{Y}$, which has been discussed in [15].

For the PARAFAC model $[\mathcal{Z}]_3 = (\bar{\mathbf{A}}_R \odot \bar{\mathbf{A}}_T) \bar{\mathbf{G}}^T$, its essential uniqueness condition is given by

$$k(\bar{\mathbf{A}}_R) + k(\bar{\mathbf{A}}_T) + k(\bar{\mathbf{G}}) \geq 2K + 2. \quad (54)$$

The k -rank of $\bar{\mathbf{G}}$ is directly linked to the k -rank of $\mathbf{G} = \mathbf{G}_d \text{diag}(\alpha) = [\alpha_1 \mathbf{g}_1^{\text{dop}}, \dots, \alpha_K \mathbf{g}_K^{\text{dop}}]$. Since $\mathbf{G}_d$ has a Vandermonde structure with column vectors $\mathbf{g}_k^{\text{dop}}$ for $k = 1, \dots, K$ and $f_d^{k_1} \neq f_d^{k_2}$, it is full k -rank [42]. Each column of $\mathbf{G}$ can be expressed as $\alpha_k \mathbf{g}_k^{\text{dop}}$, and α_k is a randomly generated coefficient. Obviously $\mathbf{G}$ is a columnwise-scaled Vandermonde matrix, and still follows the full k -rank nature [7]. Besides, it is reasonable to suppose that $N \geq K$ for time-varying mmWave channels, and thus $k(\mathbf{G}) = k(\mathbf{G}_d) = \min(N, K) = K$. Since $\bar{\mathbf{G}} = \mathbf{\Gamma}_{2N}^T [\mathbf{G}^T, \mathbf{G}^H \mathbf{\Pi}_N]^T$, we further know $k(\bar{\mathbf{G}}) \geq K$. We then focus on the k -rank of $\bar{\mathbf{A}}_R = \mathbf{\Gamma}_{N_{\text{RF}}}^H \mathbf{W}^H \mathbf{A}_R \in \mathbb{R}^{N_{\text{RF}} \times K}$ and $\bar{\mathbf{A}}_T = \mathbf{\Gamma}_R^H \mathbf{F}^T \mathbf{A}_T^* \in \mathbb{R}^{R \times K}$. Since $\mathbf{W}$ and $\mathbf{F}$ are randomly generated, they follow the full k -rank nature. In addition, each column of $\mathbf{A}_R$ and $\mathbf{A}_T$ are independent of other columns since $\theta_{k_1} \neq \theta_{k_2}$, $\phi_{k_1} \neq \phi_{k_2}$. Hence $\bar{\mathbf{A}}_R$ and $\bar{\mathbf{A}}_T$ are full k -rank, which means $k(\bar{\mathbf{A}}_R) = \min(N_{\text{RF}}, K) = K$ and $k(\bar{\mathbf{A}}_T) = \min(R, K) = K$. Recalling the Kruskal's condition (54), essential uniqueness is guaranteed if

$$\min(N_{\text{RF}}, K) \geq 2. \quad (55)$$

It holds even in the case of $K \geq N_{\text{RF}} = 2$, which means that the proposed OR-TALS algorithm can guarantee the identifiable estimation with $K \geq 2$ scattering paths even when MS is only equipped with two RF chains.

Note that the Kruskal's constrained conditions (52) and (54) are not only sufficient but also necessary for the case $R, K = 2$ or 3 [43]. If R and K , i.e., the rank of $\mathcal{Y}$ and $\mathcal{Z}$, are greater than 3, then the Kruskal's sufficient conditions are not necessary for the uniqueness of the proposed ONP-ITS-ALS algorithm. In these

scenarios, the essential uniqueness derivation of PARAFAC model is still an open issue and can be further discussed [44].

B. Complexity

The dominant computationally intensive part of the proposed ONP-ITS-ALS algorithm is attributed to the LS solutions, SVP and the line search process for each iteration. The LS estimation mainly involves in the operations of Moore-Penrose pseudo-inverse and multiplications of related matrices. We first elaborate on the calculation of $[\hat{\mathcal{X}}]_3 = [(\hat{\mathbf{S}} \odot \mathbf{C})^\dagger [\mathcal{Y}]_3]^\top$. $(\hat{\mathbf{S}} \odot \mathbf{C}) \in \mathbb{C}^{PT \times R}$ takes $(T+P)R$ flops, and its Moore-Penrose pseudo-inverse requires $\mathcal{O}(PTR^2)$ flops. The multiplication of $(\hat{\mathbf{S}} \odot \mathbf{C})^\dagger$ and $[\mathcal{Y}]_3$ costs $PTNRN_{\text{RF}}$ flops. The number of flops required by other LS calculations similarly follows that of $[\hat{\mathcal{X}}]_3$. Then we commence with the required flops of the calculation of SVP. For a matrix $\mathbf{B}$ with size $I \times J$, the complexity of its SVD is $\mathcal{O}(\min(I, J)IJ)$. Therefore, $\text{SVP}([\hat{\mathcal{X}}]_3^n)$ for $n = 1, \dots, N$ costs $\mathcal{O}(NRN_{\text{RF}}^2)$ flops. For the line search process, the calculations of $\mathbf{T}_i \in \mathbb{R}^{RN_{\text{RF}} \times 2N}$ ($i = 1, 2, 3, 4$) and Δ need $8NRN_{\text{RF}}K + 4RN_{\text{RF}}K$ and $20NRN_{\text{RF}}$ flops [40], respectively.

The dominant terms of the first and second stages for each iteration have computational complexity of order $\mathcal{O}(PTR^2 + N_{\text{RF}}NPR^2 + NRN_{\text{RF}}^2) + 2PTNRN_{\text{RF}} + (2P+T+N_{\text{RF}}N)R$ and $\mathcal{O}(RN_{\text{RF}}K^2 + NRK^2 + N_{\text{RF}}NK^2) + 14NRN_{\text{RF}}K + 4RN_{\text{RF}}K + 20NRN_{\text{RF}} + (4N + 2N_{\text{RF}} + 2R)K$, respectively. Note that in the second stage, the estimation processes are based on real-valued instead of complex-valued matrices. Hence the proposed ONP-ITS-ALS algorithm has much less complexity compared with the TNP-TS-ALS joint estimation scheme.

V. SIMULATION RESULTS

In this section, exhibitions of Monte Carlo simulations are presented to show the superiority of the proposed ONP-ITS-ALS algorithm. The probability of detection (PoD), i.e., $\Pr(\hat{K} = K)$, averaged over all Monte Carlo realizations, is defined to evaluate the estimation performance of the number of channel scattering paths K . The information symbol detection performance is depicted with the bit error rate (BER) of transmitted information symbols. The normalized mean square error (NMSE) is used to characterize the time-varying channel estimation performance, and its definition is

$$\text{NMSE} = \frac{1}{MN} \left(\sum_{m=1}^M \sum_{n=1}^N \frac{\|\hat{\mathbf{H}}[n]_m - \mathbf{H}[n]_m\|_{\text{F}}^2}{\|\mathbf{H}[n]_m\|_{\text{F}}^2} \right), \quad (56)$$

where $M = 10000$ is the number of Monte Carlo runs, the subscript m represents the m -th independent Monte Carlo run, $\mathbf{H}[n]$ and $\hat{\mathbf{H}}[n]$ are real and estimated time-varying channel at the n -th time block, respectively. Channel parameters, including AoAs, AoDs, Doppler shifts and complex path gains, can also be retrieved by the proposed ONP-ITS-ALS algorithm. The mean square error (MSE) of any parameter β is introduced to evaluate the channel parameters estimation performance [7], and it is

expressed as

$$\text{MSE}(\beta) = \frac{1}{M} \left(\sum_{m=1}^M \|\hat{\beta}_m - \beta_m\|_2^2 \right). \quad (57)$$

The signal-to-noise ratio (SNR) is defined as [7], [23]

$$\text{SNR} = 10 \lg \frac{\|\mathcal{Y}^{\text{MS}} - \mathcal{N}\|_{\text{F}}^2}{\|\mathcal{N}\|_{\text{F}}^2}, \quad (58)$$

where $\mathcal{Y}^{\text{MS}}$ and $\mathcal{N}$ represent the fourth-order received signal and noise tensors, respectively. Since the first row of signal matrix $\mathbf{S}$ needs to be known for eliminating the scaling ambiguity, the transmission rate of the proposed scheme is $\frac{(T-1)R}{NTP} \log_2 \vartheta$ bits per channel use, where ϑ denotes the cardinality of the information symbol constellation.

The following system settings are taken as examples to show the joint estimation performance of the proposed ONP-ITS-ALS algorithm. In simulations, the communication system is supposed to function with carrier frequency $f_c = 28$ GHz. The spacing between two adjacent antennas is set to half the signal wavelength. 64QAM is adopted to modulate information symbols. The sampling period T_s is set to $2 \mu\text{s}$. The KRST coding matrix $\mathbf{C}$ is randomly generated by $\exp(-j2\pi u)/\sqrt{P}$, where $j = \sqrt{-1}$ and u is a random variable chosen from the i.i.d. standard uniform distribution. The complex path gain α_k is randomly generated from the complex Gaussian distribution with zero mean and unit variance. AoDs and AoAs take continuous values and are randomly distributed in $[0, 2\pi]$. The Doppler shift f_d^k for each path is uniformly chosen from $[0, f_d^{\text{max}}]$ and $f_d^{\text{max}} = 5000$ Hz. Unless otherwise specified, the ULAs are assumed to be deployed at the BS and MS.

A. Estimation Performance of the Number of Channel Scattering Paths for Different K , N , T and P

In the first example, the estimation performance of the number of channel scattering paths K is evaluated for different measurements. PoDs curves versus SNR for different reasonable values of K are plotted in Fig. 2. Default system parameters are set as: $M_T = N_R = 64$, $M_{\text{RF}} = N_{\text{RF}} = R = 8$, $N = 30$ and $P = 10$. It can be seen from Fig. 2 that the PoDs increase monotonically with SNR and attain 1 at sufficiently high SNR for all reasonable number of scattering paths K . The histograms of estimated number of K for different RF chains N_{RF} , time blocks N , and coding length P are shown in Fig. 3 at SNR = 20 dB with $K = 3$. As shown in Fig. 3, the number of correct estimates of K is much greater than that of incorrect estimates of K over all 10000 Monte Carlo simulations. It is concluded from the simulations that the number of channel scattering paths K can be perfectly estimated when SNR is high, e.g., SNR = 20 dB. In order to better evaluate the joint signal and channel parameter estimation performance, we reasonably assume that the value of K is perfectly estimated or known a prior in the following examples.

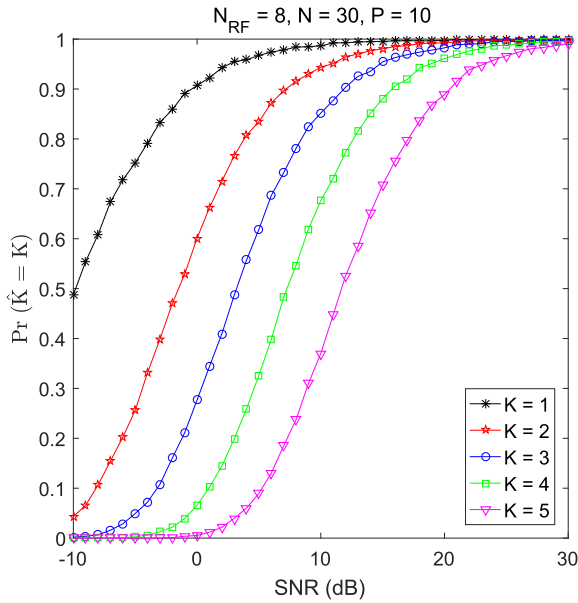


Fig. 2. PoD versus SNR for different K .

B. Comparison With TNP-Based Algorithms

TNP-based Algorithm: For the TNP-TS-ALS algorithm, signal matrix $\mathbf{S}$ and compound matrix $[\mathcal{X}]_3$ are jointly estimated in the first stage by the BALS algorithm. While for the TNP-SVD-ALS algorithm, the SVD-based algorithm [32], [33] is adopted for joint estimation in the first stage. For both TNP-based algorithms, the estimated PARAFAC tensor model $\hat{\mathcal{X}}$ is directly used to estimate channel parameters via the TALS algorithm in the second stage.

Compared with these TNP-based algorithms, the proposed ONP-ITS-ALS algorithm makes full use of the sparse scattering nature of mmWave channels in the first stage. In the second stage, we first convert the complex inner PARAFAC model into a real one by using the forward-backward and unitary transformation technique. Then the TALS algorithm is optimized by the line search processes to accelerate the fitting of the inner PARAFAC model.

In the second example, the joint channel parameter estimation and information symbol detection performance of the proposed ONP-ITS-ALS algorithm is compared with that of TNP-TS-ALS and TNP-SVD-ALS algorithms, in terms of symbol BER, channel NMSE and channel parameters MSEs. In particular, we supply with illustrations of iteration number and mean processing time to reflect the computational complexity of these three algorithms. The signal BER curve based on the estimated CSI of the proposed ONP-ITS-ALS algorithm is also plotted in this example, marked as “ONP-ITS-ALS, reEst”. In this case, the compound matrix $[\mathcal{X}]_3$ is first reconstructed by estimated channel parameters, and then the signal matrix $\mathbf{S}$ is estimated by LS with (28). Besides, the BER, NMSE and MSEs performance of the proposed ONP-ITS-ALS algorithm under the typical two-dimensional array geometry UPAs are also investigated in this example. In order to ensure the identifiable estimation of the TNP-SVD-ALS algorithm, the coding length P must be greater than the number of data streams R . Therefore, $P = 10 > R = 8$

TABLE III
ITERATION NUMBER OF TALS PART, ULAS CONFIGURATION

SNR (dB)	-10	-7	-4	-1	2	5	8	11	14	17	20
ONP-ITS-ALS	13	9	10	11	10	11	9	10	10	9	12
TNP-TS-ALS	130	137	157	179	172	183	197	183	186	205	201
TNP-SVD-ALS	113	119	125	138	146	163	175	180	182	198	232

is chosen in this example. Other system parameters are set as: $M_T = N_R = 64$ for ULAs, $M_T^x = M_T^y = N_R^x = N_R^y = 8$ for UPAs, $M_{RF} = N_{RF} = 8$, $K = 3$, $N = 30$ and $T = 8$.

The BER, NMSE, and MSEs curves of these three joint estimation algorithms are plotted in Fig. 4 and Fig. 5, respectively. For these three joint estimation algorithms, they jointly estimate signal matrix $\mathbf{S}$ and compound matrix $[\mathcal{X}]_3$ in the first stage, and estimate channel parameters in the second stage. The iteration number of TALS part in the second stage is shown in Table III. The mean processing time in the first stage and second stage, respectively marked as “F-S” and “S-S” for convenience, of all these algorithms are displayed in Fig. 6. In the first stage, due to the SVP optimization, the BALS mean processing time of the proposed ONP-ITS-ALS algorithm is slightly higher, but its BER performance is better than the other two competitive algorithms. The BER difference will become more pronounced with increasing number of N_{RF} (see Fig. 7). In the second stage, the real-valued PARAFAC model is harvested by the forward-backward and unitary transformation technique. The observation data used for estimation is twice of the original, and hence the proposed ONP-ITS-ALS algorithm meliorates the channel estimation performance. At a higher SNR, the NMSE gap between ONP-ITS-ALS and TNP-TS-ALS algorithms is around 1 dB. More favorable performance improvement appears at low SNR. The trend of MSEs curves of these three algorithms is consistent with that of NMSE curves. It can also be seen from Fig. 4 that great estimated CSI improves the symbol detection performance. Moreover, the proposed ONP-ITS-ALS algorithm shows the lowest TALS mean processing time, which comes from the fact that its iteration number required for the TALS part is significantly less than the other two competitive algorithms due to the line search processes. In addition, the inverse matrix calculation is the dominant computationally intensive part for ALS. Compared with traditional complex-valued PARAFAC model, the derived real-valued PARAFAC model saves a lot of calculations when computing the inverse matrix.

The BER, NMSE, MSEs curves of the proposed algorithm with UPAs configuration are also plotted in Fig. 4 and Fig. 5. It can be seen that when the antenna layout is changed from a linear array to a planar array, the proposed ONP-ITS-ALS algorithm still shows effective and great joint estimation performance.

C. Impact of N_{RF}

In the third example, we are concerned about the impact of increasing N_{RF} on the joint estimation performance. System parameters are set as: SNR = 0 dB, $M_T = 64$, $N_R = 32$, $M_{RF} = R = 8$, $K = 3$, $N = 40$, $T = 12$ and $P = 12$.

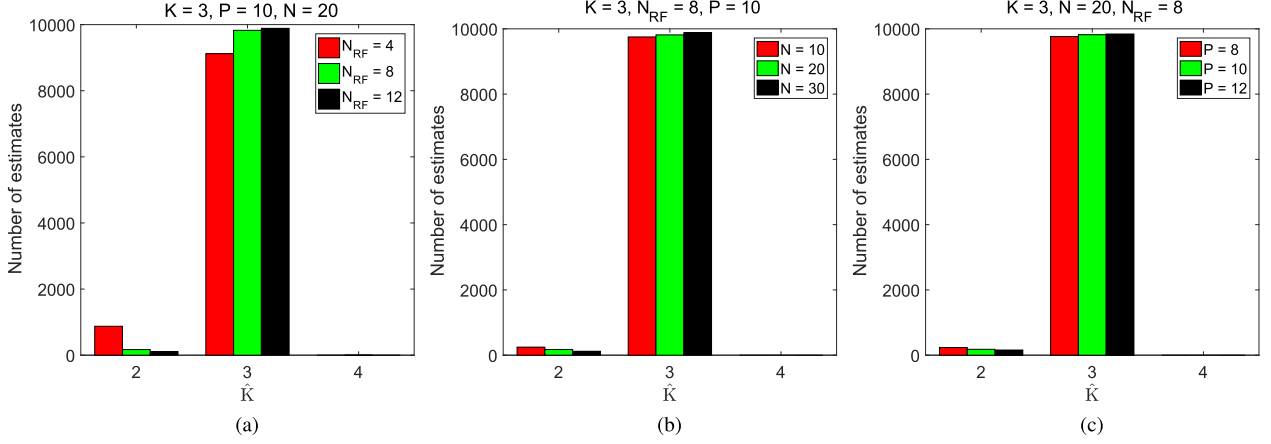
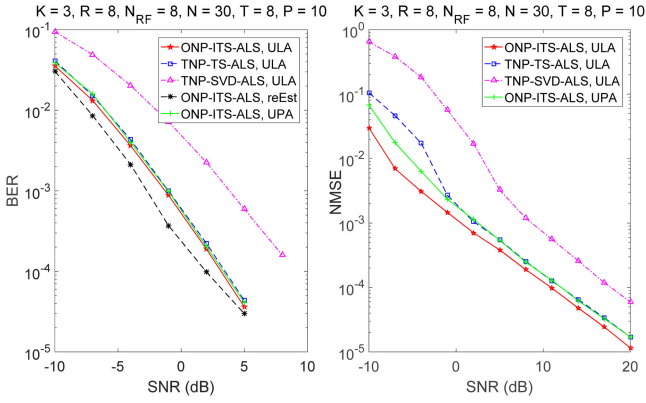
Fig. 3. Histogram of estimated number of K for different N_{RF} , N and P at SNR = 20 dB.

Fig. 4. BER and NMSE versus SNR compared with TNP-based algorithms.

The BER and NMSE performances versus N_{RF} of ONP-ITS-ALS, TNP-TS-ALS and TNP-SVD-ALS algorithms are depicted in Fig. 7. It can be seen that with the increment of N_{RF} , the BER gap between the proposed ONP-ITS-ALS and the other two algorithms tends to be larger, while the NMSE gap tends to be stable. The proposed ONP-ITS-ALS algorithm makes full use of the low-rank structure of mmWave channel for symbol detection in the first stage. When N_{RF} increases, its utility becomes prominent. As a result, its BER performance improvement is significant in this case. Also, the compound matrix $[\mathcal{X}]_3$ estimation becomes better. However, N_{RF} not only affects the estimation accuracy of signal $\mathbf{S}$ and compound matrix $[\mathcal{X}]_3$, but also controls the size of the inner PARAFAC model used for channel estimation. Increasing N_{RF} is conducive to the OR-TALS performance improvement based on the inner PARAFAC model. Therefore, the NMSE performance is subject to N_{RF} , and results in a gradually narrowing gap between the proposed ONP-ITS-ALS and the other two algorithms.

D. Impact of N , T , P and K

In the fourth example, the NMSE performance of the proposed ONP-ITS-ALS algorithm with increasing number of measurements is compared with that of the TNP-TS-ALS algorithm in Fig. 8. According to the first two experiments, the joint estimation performance of TNP-SVD-ALS is apparently worse

than the other two algorithms. Besides, the TNP-SVD-ALS algorithm can not guarantee the unique determination of $\mathbf{S}$ and $[\mathcal{X}]_3$ in the case of $P < R$. Therefore, it is not included in this experiment.

1) *Impact of N , T and P :* NMSE curves versus time blocks N , time slots T , and coding length P for different SNR are plotted in Fig. 8(a), Fig. 8(b) and Fig. 8(c), respectively. Default system parameters are set as: $M_T = 64$, $N_R = 32$, $M_{\text{RF}} = R = 8$, $N_{\text{RF}} = 4$ and $K = 3$.

In Fig. 8(a), N varies from 3 to 160, $T = 10$ and $P = 10$. In Fig. 8(b), T varies from 2 to 20, $N = 40$ and $P = 10$. In Fig. 8(c), P varies from 2 to 20, $N = 40$ and $T = 10$. As expected, the proposed ONP-ITS-ALS algorithm shows higher estimation accuracy for different SNR and measurements number. In Fig. 8(a), the NMSE performance of both algorithms is enhanced exponentially as N becomes large. This is attributed to the fact that the time-varying channel information is captured by the inner PARAFAC model from the aspect of time blocks. We can also find that the proposed ONP-ITS-ALS algorithm approaches the performance ceiling when N is around 40, which is only about half of the TNP-TS-ALS algorithm. Since the proposed ONP-ITS-ALS algorithm converts the inner PARAFAC model from complex-valued to real-valued, the observation date used for channel estimation is twice of the original. The result exhibited in Fig. 8(a) is roughly in line with our analysis. In Fig. 8(c), both algorithms suffer from performance loss in the case of $P = 2$, but only a slight increase of measurements number of P can yield a substantial performance melioration. It can also be seen from Fig. 8(b) and Fig. 8(c) that the channel estimation accuracy improvement of both algorithms tends to low levels when T and P are greater than 10. Therefore, we can choose appropriate parameters (for example, choose $6 \leq T \leq 10$ and $8 \leq P \leq 12$ in this experiment) to save resources and improve estimation performance to the greatest extent in practice.

2) *Impact of K :* NMSE curves versus reasonable number of scattering paths K of the proposed ONP-ITS-ALS algorithm is compared with that of the TNP-TS-ALS algorithm. As shown in the measurements of literature [21], K is set to vary from 1 to 5. Other system parameters are set as: $M_T = 64$, $N_R = 32$, $M_{\text{RF}} = N_{\text{RF}} = R = 8$, $N = 20$, $T = 8$ and $P = 10$.

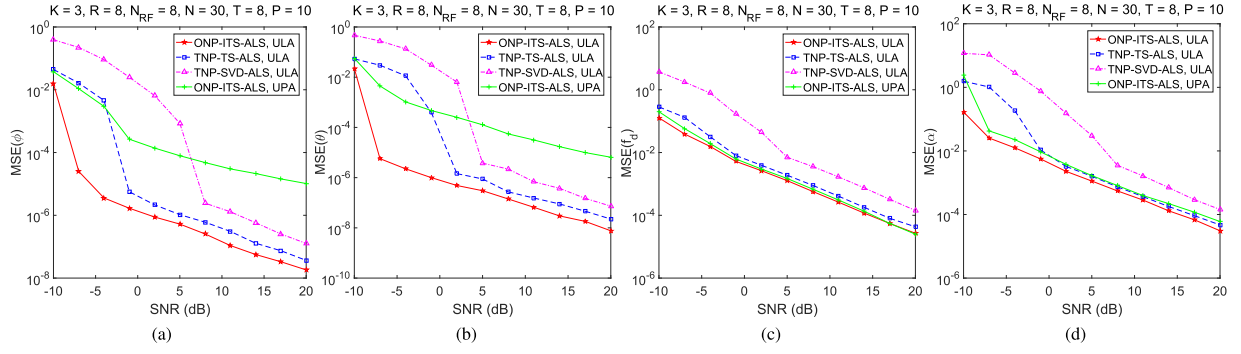


Fig. 5. MSEs versus SNR compared with TNP-based algorithms.

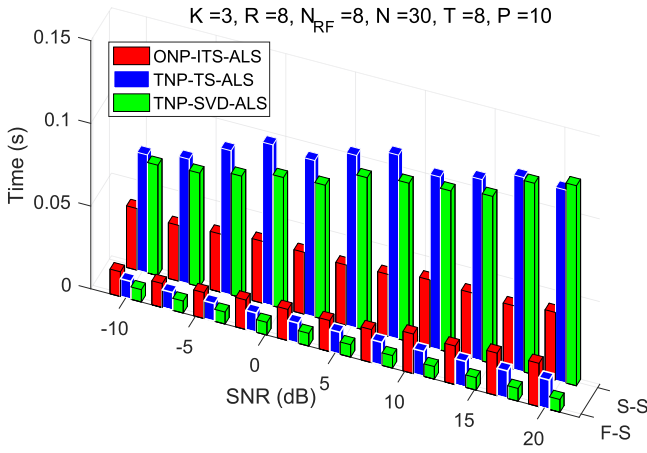
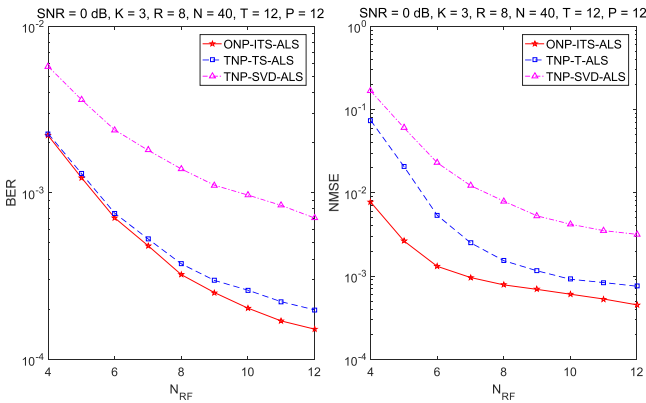


Fig. 6. Mean processing time versus SNR compared with TNP-based algorithms, ULAs configuration.

Fig. 7. BER and NMSE versus N_{RF} compared with TNP-based algorithms.

It can be seen from Fig. 8(d) that in contrast with the TNP-TS-ALS algorithm, the proposed ONP-ITS-ALS scheme exhibits enhanced channel estimation performance for different SNR and K . Note that the TNP-TS-ALS algorithm suffers from more severe NMSE performance degradation as K increases, while the proposed ONP-ITS-ALS algorithm works robustly across different scattering paths. Therefore, the proposed ONP-ITS-ALS algorithm is more attractive than the TNP-TS-ALS algorithm.

E. Comparison With Pilot-Assisted Algorithms

In the fifth example, the joint estimation performance of the proposed ONP-ITS-ALS algorithm is compared with that of the pilot-assisted ACS [5], orthogonal matching pursuit (OMP) [16], SR [6] and bilinear generalized approximate message passing like (BigAMP-like) [17], [18] algorithms in terms of BER, NMSE and mean processing time. These algorithms have great channel estimation performance when used to static channels. In order to make them suitable for time-varying channels, we apply these algorithms for channel estimation in each time block, excluding the BigAMP-like algorithm. Then, the compound matrix $[\mathcal{X}]_3$ can be constructed by estimated time-varying channels. Finally, the signal matrix $\mathbf{S}$ is estimated by LS with (28). If the value of N is large, these competitive algorithms will suffer from high computational complexity. Therefore, $N = 6$ is chosen for this experiment, which not only satisfies the identifiability conditions of the proposed ONP-ITS-ALS algorithm, but also does not excessively increase the computational burden of these competitive algorithms. Other system parameters are set as: $M_T = N_R = 64$, $M_{RF} = N_{RF} = R = 8$, $K = 3$, $T = 8$ and $P = 10$.

The ACS algorithm detects AoDs/AoAs with S stages. The number of training precoding and combining vectors per stage is $K_{BS} = K_{MS} = 2$. In particular, two different grids ($G_{BS} \times G_{MS}$) are used to quantize AoDs and AoAs in this experiment. The first one is $G_{BS} = G_{MS} = 384$ (marked as “coarse”), and the second one is $G_{BS} = G_{MS} = 1536$ (marked as “fine”). The required number of stages is equal to $S = \max(\log_{K_{BS}}(G_{BS}/K), \log_{K_{MS}}(G_{MS}/K))$. For the OMP algorithm, $Q = 20$ frames are utilized to transmit pilots and pilots length is set as 96. For the BigAMP-like algorithm, pilots length is set as 224. Long pilots length makes the SR algorithm computationally expensive. Therefore, we set pilots length as 32 to yield an accurate channel estimation without prohibitive cost.

The BER and NMSE performance of these algorithms are shown in Fig. 9. In particular, the BER curve with perfect CSI is also plotted for reference. In order to give an exhibition of the computational complexity comparison, we supply with the mean processing time in Fig. 10. It can be seen that both BER and NMSE performance of the proposed ONP-ITS-ALS algorithm are superior to that of OMP, as well as in terms of computational cost. In the low SNR regime,

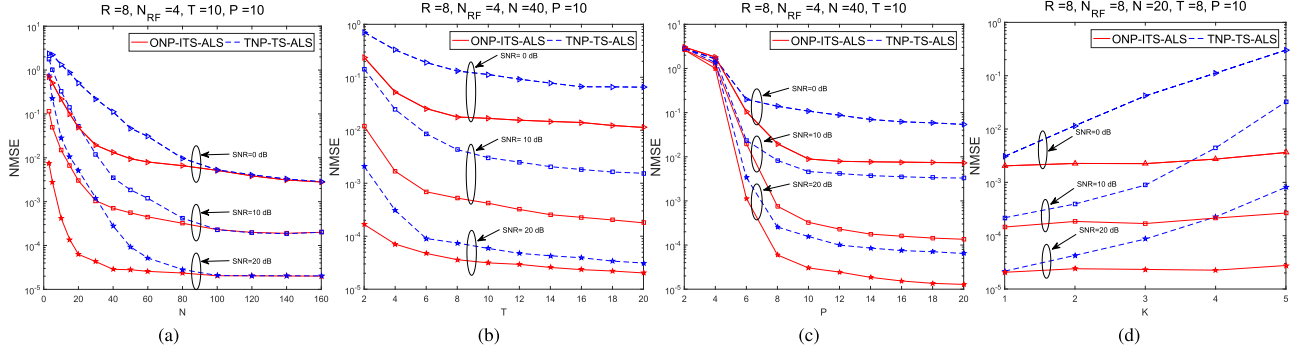
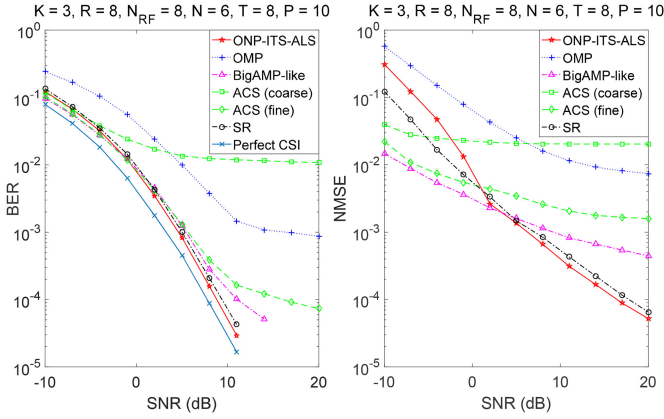

 Fig. 8. NMSE versus N , T , P and K for different SNR.


Fig. 9. BER and NMSE versus SNR compared with pilot-assisted algorithms.

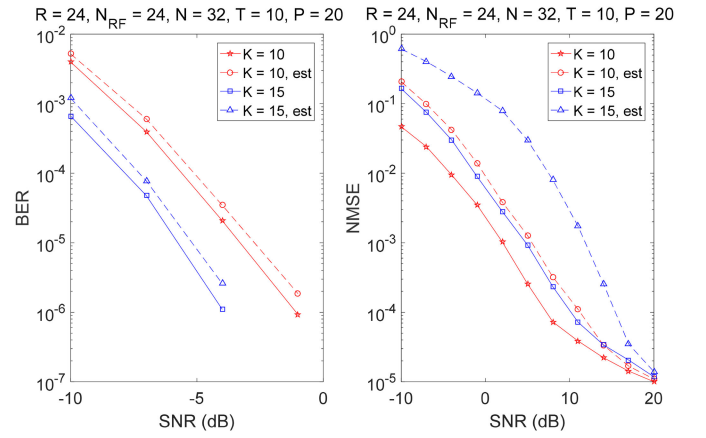
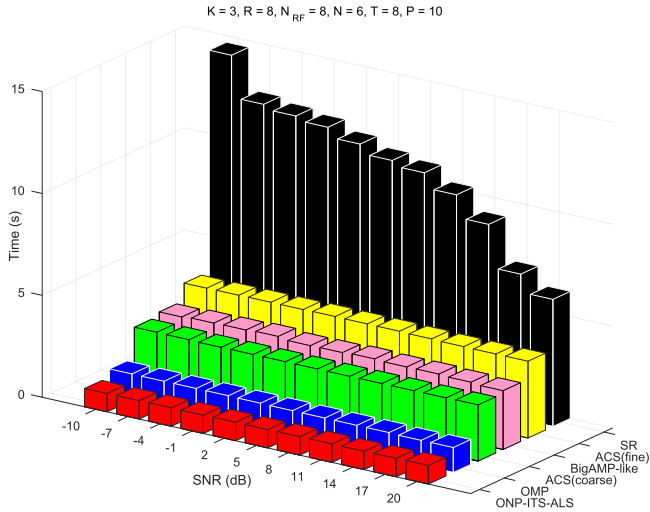

 Fig. 11. BER and NMSE versus SNR for the estimated K and large K .


Fig. 10. Mean processing time versus SNR compared with pilot-assisted algorithms.

the BER performance of the proposed ONP-ITS-ALS algorithm is similar to that of BigAMP-like, ACS, and SR algorithms, and the NMSE performance is slightly inferior to others. While in the case of high SNR, the proposed ONP-ITS-ALS algorithm achieves higher signal and channel estimation accuracy than

others. Meanwhile, its meaning processing time cost is significantly lower than others. Besides, the proposed ONP-ITS-ALS algorithm saves spectral resources.

F. Joint Estimation Performance for the Estimated K and Large K

In the final example, we are interested in the impact of the estimated K on the joint estimation performance of the proposed ONP-ITS-ALS algorithm. The cases with a large number of channel scattering paths, i.e., $K = 10$ and $K = 15$, are also considered. Other system parameters are set as: $M_T = N_R = 128$, $M_{RF} = N_{RF} = R = 24$, $N = 32$, $T = 10$ and $P = 20$.

It can be seen from Fig. 11 that the proposed ONP-ITS-ALS scheme still works well for the estimated K and large K . With the increase of K , the BER performance is significantly improved, while the NMSE performance suffers from degradation due to increased channel parameters to be estimated. In the low-to-moderate SNR regime, since it is difficult to estimate the value of K accurately and properly, the BER and NMSE performance for estimated K are worse than that for perfect K . As SNR increases, the probability of correctly estimating K increases. Therefore, the gap of NMSE curves between estimated K and perfect K gradually decreases.

VI. CONCLUSION

In this paper, we have proposed an ONP-based joint estimation algorithm for the mmWave massive MIMO time-varying channel. Compared with competitive TNP-based and pilot-assisted estimation schemes, the proposed ONP-ITS-ALS algorithm shows superior BER, NMSE and MSEs performance, as well as great robustness. Meanwhile, its computational complexity and pilot overhead are greatly reduced. Perspective of our work contains an extension to multi-user mmWave massive MIMO-OFDM systems for joint channel parameters estimation with dual-wideband effects [45] under unknown number of scatter paths and incomplete information. By taking advantage of the internal relationship between the received signals at each subcarrier, the fifth-order tensor model will be considered to be constructed. The accurate determination of high-order tensor rank is quite difficult especially in the presence of missing entries, which is more challenging. Novel tensor fitting algorithms such as the Bayesian inference based algorithm [46] will be developed to address these issues, and associated CRB will also be derived in that work.

APPENDIX

In this section, the real-valued PARAFAC model are deduced in typical UPAs without limitation of phase reference. According to formulas (6)–(9), we have

$$\mathbf{A}_{\text{UPA}} = \mathbf{A}_{\text{ULA}}^y \odot \mathbf{A}_{\text{ULA}}^x, \quad (59)$$

where $\mathbf{A}_{\text{ULA}}^b$ with $b = x, y$ are defined by (9) with $\mathbf{a}_k$ replaced by $\mathbf{a}_{\text{ULA}}(\eta^b)$ that follows the form of (6). Although the conjugate centrosymmetric property is lost in this case, $[\mathcal{X}]_1$ can also be converted into a real-valued PARAFAC model. Define

$$\begin{aligned} \Xi = & \text{diag}(e^{-j2\pi d \left(\frac{(M^x-1)\eta_1^x + (M^y-1)\eta_1^y}{2} \right) / \lambda}, \\ & \dots, e^{-j2\pi d \left(\frac{(M^x-1)\eta_K^x + (M^y-1)\eta_K^y}{2} \right) / \lambda}, \end{aligned} \quad (60)$$

and

$$\tilde{\mathbf{A}}_{\text{UPA}} = \tilde{\mathbf{A}}_{\text{ULA}}^y \odot \tilde{\mathbf{A}}_{\text{ULA}}^x, \quad (61)$$

where $\tilde{\mathbf{A}}_{\text{ULA}}^b$ with $b = x, y$ are defined by (9) with $\mathbf{a}_k$ replaced by $\tilde{\mathbf{a}}_{\text{ULA}}^b(\eta_k^b)$ that follows the form of (34). $\mathbf{A}_{\text{UPA}}$ satisfies the following characteristics

$$\begin{cases} \Pi_M \mathbf{A}_{\text{UPA}}^* = \tilde{\mathbf{A}}_{\text{UPA}} \Xi^* \\ \mathbf{A}_{\text{UPA}} = \tilde{\mathbf{A}}_{\text{UPA}} \Xi \end{cases}. \quad (62)$$

Further, $\mathbf{A}_{\text{TR}} = \mathbf{A}_{\text{R}} \odot \mathbf{A}_{\text{T}}^*$ also satisfies the characteristics of (62), i.e.,

$$\begin{cases} \Pi_{M_T N_R} \mathbf{A}_{\text{TR}}^* = \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}}^* \\ \mathbf{A}_{\text{TR}} = \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}} \end{cases}, \quad (63)$$

where $\Xi_{\text{TR}} = \Xi_{\text{R}} \Xi_{\text{T}}^*$, $\tilde{\mathbf{A}}_{\text{TR}} = \tilde{\mathbf{A}}_{\text{R}} \odot \tilde{\mathbf{A}}_{\text{T}}^*$, Ξ_b and $\tilde{\mathbf{A}}_b$ with $b = T, R$ are defined by (60) and (61), respectively.

We rewrite $[\mathcal{X}]_1$ and $\Pi_{RN_{\text{RF}}}[\mathcal{X}]_1^* \Pi_N$ as

$$\begin{aligned} [\mathcal{X}]_1 &= (\mathbf{W}^H \otimes \mathbf{F}^T) \mathbf{A}_{\text{TR}} \mathbf{G}^T \\ &= (\mathbf{W}^H \otimes \mathbf{F}^T) \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}} \mathbf{G}^T, \end{aligned} \quad (64)$$

$$\begin{aligned} \Pi_{RN_{\text{RF}}}[\mathcal{X}]_1^* \Pi_N &= \Pi_{RN_{\text{RF}}} (\mathbf{W}^T \otimes \mathbf{F}^H) \mathbf{A}_{\text{TR}}^* \mathbf{G}^H \Pi_N \\ &\stackrel{(a)}{=} \Pi_{RN_{\text{RF}}} (\mathbf{W}^T \otimes \mathbf{F}^H) \Pi_{M_T N_R} \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N \\ &\stackrel{(b)}{=} (\Pi_{N_{\text{RF}}} \otimes \Pi_R) (\mathbf{W}^T \otimes \mathbf{F}^H) (\Pi_{N_R} \otimes \Pi_{M_T}) \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N \\ &\stackrel{(c)}{=} (\Pi_{N_{\text{RF}}} \mathbf{W}^T \Pi_{N_R} \otimes \Pi_R \mathbf{F}^H \Pi_{M_T}) \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N \\ &\stackrel{(d)}{=} (\mathbf{W}^H \otimes \mathbf{F}^T) \tilde{\mathbf{A}}_{\text{TR}} \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N, \end{aligned} \quad (65)$$

where (a) follows from the property (63), (b) follows from $\Pi_{RN_{\text{RF}}} = \Pi_{N_{\text{RF}}} \otimes \Pi_R$ and $\Pi_{M_T N_R} = \Pi_{N_R} \otimes \Pi_{M_T}$, (c) follows from the property (1), and (d) follows from the formula (33). Therefore, the centro-Hermitian matrix $\bar{\mathbf{Z}} = [[\mathcal{X}]_1, \Pi_{RN_{\text{RF}}}[\mathcal{X}]_1^* \Pi_N]$ can be simplified as

$$\begin{aligned} \bar{\mathbf{Z}} &= (\mathbf{W}^H \otimes \mathbf{F}^T) \tilde{\mathbf{A}}_{\text{TR}} [\Xi_{\text{TR}} \mathbf{G}^T, \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N] \\ &= (\mathbf{W}^H \tilde{\mathbf{A}}_{\text{R}} \odot \mathbf{F}^T \tilde{\mathbf{A}}_{\text{T}}^*) [\Xi_{\text{TR}} \mathbf{G}^T, \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N] \\ &= (\tilde{\mathbf{Q}} \odot \tilde{\mathbf{P}}) [\Xi_{\text{TR}} \mathbf{G}^T, \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N]. \end{aligned} \quad (66)$$

The real-valued PARAFAC mode-3 matrix $[\mathcal{Z}]_3$ is then obtained by

$$\begin{aligned} [\mathcal{Z}]_3 &= (\Gamma_{N_{\text{RF}}}^H \otimes \Gamma_{\text{R}}^H) \bar{\mathbf{Z}} \Gamma_{2N} \\ &= \left((\Gamma_{N_{\text{RF}}}^H \tilde{\mathbf{Q}}) \odot (\Gamma_{\text{R}}^H \tilde{\mathbf{P}}) \right) [\Xi_{\text{TR}} \mathbf{G}^T, \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N] \Gamma_{2N} \\ &= (\bar{\mathbf{A}}_{\text{R}} \odot \bar{\mathbf{A}}_{\text{T}}) \bar{\mathbf{G}}^T, \end{aligned} \quad (67)$$

where $\bar{\mathbf{A}}_{\text{R}} = \Gamma_{N_{\text{RF}}}^H \tilde{\mathbf{Q}}$, $\bar{\mathbf{A}}_{\text{T}} = \Gamma_{\text{R}}^H \tilde{\mathbf{P}}$ and $\bar{\mathbf{G}}^T = [\Xi_{\text{TR}} \mathbf{G}^T, \Xi_{\text{TR}}^* \mathbf{G}^H \Pi_N] \Gamma_{2N}$ are real-valued matrix factors.

ULAs can be regarded as a special case of UPAs, where all antennas are placed in the same direction, e.g., x-direction or y-direction. Therefore, the derivation of the real-valued PARAFAC model in typical ULAs without limitation of phase reference is omitted.

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